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Research
Theory and Key Technologies in Satellite Internet Networking—Review

Network-Layer Perspectives on Satellite–Terrestrial Integrated Networks in 6G: A Comprehensive Review

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ABSTRACT

Satellite–terrestrial networks have garnered significant attention in recent years and are extensively applied in intelligent transportation and emergency rescue. This paper provides a comprehensive review of the latest research advancements in satellite–terrestrial integrated network (STIN) technologies from a network perspective, dividing STIN technologies into three categories according to network service flows—namely, topology maintenance, network routing, and orchestration transmission technologies. Furthermore, a novel network-layer perspective is considered to examine the applications of STINs across various domains, along with related frameworks, platforms, simulators, and datasets. Finally, this paper explores the mainstream research directions in STIN technologies, with an innovative focus on the network layer. It reviews the existing literature, outlines future trends, and discusses opportunities for collaboration with related fields.

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1. Introduction

1.1. Background

With the rapid advancements in fifth-generation (5G) and sixth-generation (6G) technologies, along with aerospace innovations, satellite–terrestrial integrated networks (STINs) have garnered widespread attention. These networks are intended to expand the application range of terrestrial systems while addressing the growing demand for ubiquitous connectivity. STINs have several key attributes, including enhanced coverage, low-latency communication, and improved resilience. By leveraging satellite networks, STINs provide seamless communication and data transmission across diverse environments. This approach can effectively bridge the digital divide to ensure connectivity for four billion people worldwide [1]. Notably, satellite networks are essential for the shift from 5G to 6G, making 6G an evolution of 5G enhanced by satellite connectivity.

Satellite networks are increasingly utilized in various domains, including remote sensing and telecommunications. In particular, they support applications such as environmental monitoring, location services, direct access to smartphones, and distributed computing [2]. A key reason for using satellite networks is their capacity for complex tasks, such as multi-participant data processing. Satellite networks also offer greater scalability and resilience than ground networks, especially in remote areas or during failures [3]. Thanks to these features, satellite networks can be employed to adapt to various scenarios, such as rural coverage.

Interestingly, terrestrial networks also play a crucial role in supporting satellite networks by handling high-speed data processing and local connectivity. Terrestrial networks have transitioned from traditional infrastructure with limited coverage to a vital part of global connectivity, managing tasks such as satellite–terrestrial data transmission and coordinating satellite handovers.

To some extent, satellite networks can be seen as an extension and expansion of terrestrial networks. Traditionally, emergency communication relies on mobile communication, such as unmanned aerial vehicles (UAVs) and other vehicles. However, these facilities may be compromised or rendered inoperative in remote areas or during emergency and disaster relief. To enhance emergency rescue, satellite networks are used for their wide

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coverage, which enables reliable global communication in remote or disaster-hit areas. Fig. 1 illustrates how a space–air–ground integrated network (SAGIN) offers protection to cities.

1.2. Motivation

Previous research has primarily emphasized the physical layer of satellite networks, including signal transmission, modulation, and channel models [4–7]. Building on this foundation, this review explores satellite networks from a network-layer perspective. Although studies have been conducted on the network layer [8,9], a comprehensive analysis of network technologies in satellite communications is still lacking, and the literature on this topic is in its infancy. Existing reviews do categorize technologies in STINs, but they often lack updated insights, consistent classification criteria, and coverage of key topics. To address these gaps, this study summarizes relevant research, focusing on key technologies such as topology maintenance, network routing, and orchestration transmission, which demonstrate the benefits of satellite–terrestrial integration.

The integration of these technologies with non-terrestrial networks—including satellites, high-altitude platforms, and UAVs—provides the potential for global coverage. Moreover, this field is witnessing a growing influence of artificial intelligence (AI)-powered solutions. From the perspective of signal processing, AI is being utilized to optimize spectrum sensing and improve network performance in 6G networks [10]. Regarding underlying technologies, reconfigurable intelligent surfaces (RISs) integrated with AI-driven spectrum learning are emerging in satellite networks, with the aim of enhancing energy efficiency and communication performance [11]. Communication strategies also benefit from AI, as AI-assisted medium access control for RIS plays a crucial role in STINs by optimizing spectrum usage and strengthening communication quality [12]. Additionally, federated spectrum learning for RIS-aided networks enables collaborative optimization among network nodes, significantly boosting spectrum efficiency [13]. From an application perspective, AI-based multi-task learning has proven effective in enhancing coordination between UAVs and satellites, improving aerial–terrestrial communication [14]. AI-driven edge intelligence is also being explored for advanced autonomous systems, such as self-driving vehicles, and holds significant potential to advance wireless systems within the STIN framework [15]. However, establishing a stable and sustainable STIN presents several challenges. First, compared with ground-based network nodes, it is more challenging for satellites to maintain stable connections due to satellites' high mobility. Second, real-time perception and construction of network topology are necessary for collaboration among network nodes. Third, inefficient routing under various constraints can compromise data security and reduce the quality of data transmission. Fourth, improving the transmission efficiency in satellite networks requires coordination between satellites and ground stations. Thus, determining how to address the above challenges has become a crucial step in accelerating the application of satellite networks.

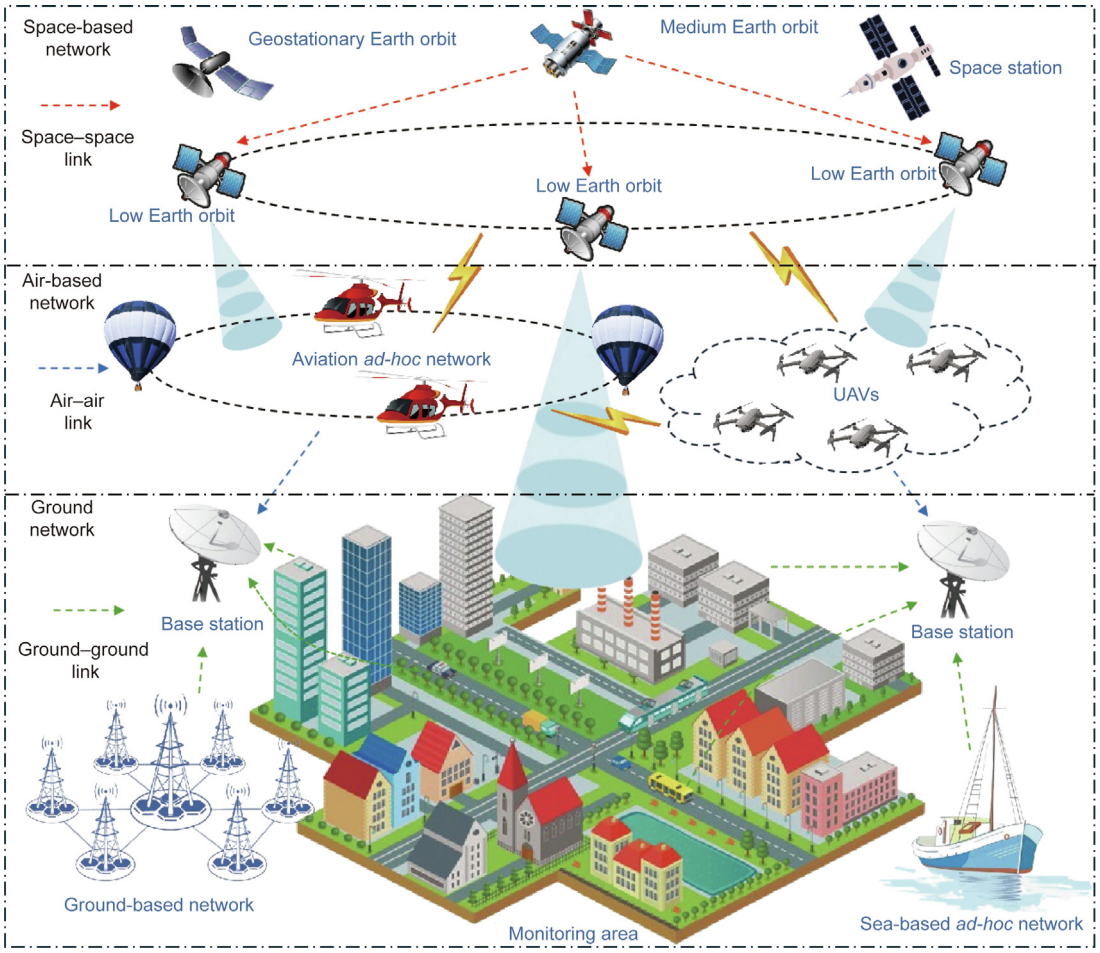


Fig. 1. An illustration of SAGIN providing comprehensive city protection.

1.3. Contribution

This survey explores STIN technologies and presents a novel focus on the network layer that has not been extensively addressed in previous surveys, highlighting SAGIN as a mature and established model in the field. It focuses on optimizing transmission paths, increasing data-transmission efficiency, and improving overall network performance. Based on these considerations, the survey proposes major network-layer challenges along with three technology classifications to address them—namely, topology maintenance, network routing, and orchestration transmission. Furthermore, it presents the latest developments in satellite networks, including platforms and simulators, while exploring future research directions and prospects. A comparison with existing relevant reviews is illustrated in Table 1 [6,8,16–22].

The contributions of this survey are summarized as follows:

(1) This work aims to provide a deeper analysis of key technologies in STINs from a network-layer perspective, focusing on topology maintenance, network routing, and orchestration transmission. We explore these technologies in the context of the unique characteristics of STINs, analyzing their performance, challenges, and recent advancements.

(2) We analyze innovations in satellite networks, focusing on their roles in diverse scenarios. Additionally, we investigate emerging applications in transportation, finance, and industry to offer a broader perspective on satellite network performance.

(3) We introduce an innovative exploration of interconnections within STINs by categorizing network frameworks, simulators, and datasets for satellite networks. Furthermore, we delve into the network-layer-specific challenges these technologies present and propose future research directions to overcome these obstacles, aiming to accelerate the development of STINs.

Fig. 2 outlines the structure of this survey paper. The remainder of this work is organized as follows: Section 2 covers STIN concepts, technology classifications, and the architecture of SAGIN. Section 3 addresses issues in STINs and outlines relevant technologies in this area such as graph-based methods. Section 4 explores topology maintenance technologies, such as node management, topology management, computation, and monitoring. Section 5 categorizes and summarizes network routing technologies based on different principles. Section 6 focuses on orchestration and transmission. Section 7 examines the main application scenarios of STINs and discusses practical application cases. Section 8 reviews related frameworks, platforms, simulators, and datasets. Section 9 discusses interactions with related fields and analyzes future research directions. Finally, Section 10 provides a summary of the review.

2. Preliminaries

A solid understanding of the concepts and architecture of 6G satellite communication models establishes a foundation for exploring STIN technologies. Thus, this section introduces the fundamental concepts of STINs and provides an in-depth examination of the architecture of SAGIN.

2.1. Introduction to STINs

A STIN enables satellites and terrestrial networks to complement each other. Satellites help fill coverage gaps in terrestrial networks, improving connectivity in remote or underserved areas.

(1) **Satellite networks**. Satellite networks are communication systems that utilize multiple satellites to facilitate wide-area data transmission. They are ideal for regions that are difficult to access via terrestrial infrastructure such as roads, bridges, and traditional communication networks. Advancements in aerospace technology and reduced launch costs have led to broader deployment of satellite networks, increasing reliance on them in various industries. These advancements have improved data-transmission rates, decreased latency, and extended coverage in satellite communication systems. Notably, emerging low-Earth-orbit (LEO) constellations are enhancing global Internet coverage and connectivity for underserved populations.

(2) **Terrestrial networks.** Terrestrial networks are communication systems based on ground infrastructures such as optical fibers, cables, radio waves, and microwaves. They provide high-speed Internet access and reliable communication in both urban and

Table 1
Comparison of existing relevant reviews.

Ref.	Year	Challenges and considerations addressed	Classification of technical focus areas				
			Resource management	System integration	Safety protection	Multilayer cooperation	Energy consumption
[16]	2018	System modeling, protocol optimization, resource acquisition and allocation	✓	–	✓	–	–
[17]	2020	Resource constraints, integration, real-time and reliable communications	✓	✓	✓	✓	–
[18]	2021	System optimization by blockchain, effective and secure communications	✓	✓	–	✓	–
[6]	2022	Security constraints from multilayer cooperation and open communication	–	✓	–	✓	–
[19]	2022	Cognitive spectrum use, secure communication, energy efficiency	✓	–	✓	✓	✓
[8]	2023	Resource constraints such as latency, throughput, capacity, and operation cost	–	–	✓	–	–
[20]	2024	Conflict between trust and privacy protection in the architecture	✓	–	–	✓	✓
[21]	2024	Data transmission obstructed by barriers, signal errors, signal fading	✓	✓	–	–	–
[22]	2025	Traditional communication no longer applicable for limitations such as dynamics	✓	–	✓	–	–
This study	2025	Topology maintenance, network routing, and orchestration transmission	✓	✓	✓	✓	✓

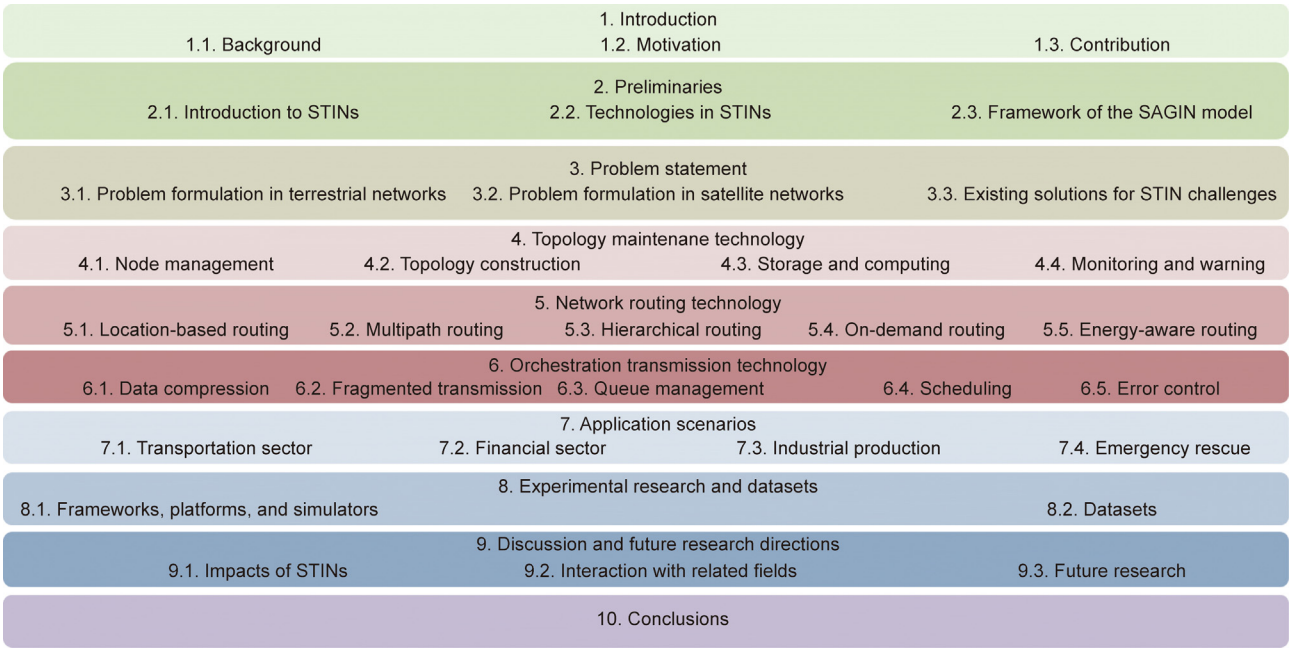


Fig. 2. The review architecture of this survey.

rural areas. Recent advancements in fiber optic and wireless technologies have made these networks essential for modern communication. Their extensive coverage and strong performance support various applications across multiple sectors.

The core concept of the STIN lies in its seamless integration of satellite and terrestrial networks, enabling them to complement and support each other to address their respective limitations. Satellite networks extend coverage to areas where terrestrial networks are unavailable, such as remote or underserved regions, ensuring global connectivity. Meanwhile, terrestrial networks enhance the overall performance by providing high-speed data transmission and ultra-low latency, particularly in densely populated urban areas where ground infrastructure can efficiently handle high traffic demands. Together, these networks create a robust communication ecosystem, ensuring continuous, reliable connectivity that meets the diverse needs of users across various environments.

2.2. Technologies in STINs

STIN technologies combine satellites with ground systems to merge spatial and terrestrial information, enhancing various applications.

(1) **Remote sensing technology.** This technology uses optical and radar imagery to improve environmental monitoring accuracy. It employs multi-temporal data fusion to analyze time-series data, enabling the monitoring of changes such as environmental shifts and agricultural growth.

(2) **Navigation technology.** This technology integrates global navigation satellite systems with ground-based augmentation systems, increasing accuracy and reliability. It also combines satellite navigation, inertial navigation, and visual navigation methods for high-precision positioning.

(3) **Communication technology.** This technology coordinates satellites with ground stations to provide wide coverage and high-bandwidth communication, benefiting remote and disaster-prone areas. It integrates satellite and terrestrial Internet for seamless global connectivity, improving overall communication network quality.

(4) **Data fusion technology.** This technology merges multi-source data from satellites and ground sources. By integrating information such as weather and environmental data, it supports decision-making. Furthermore, cloud computing and big data enable real-time processing, boosting the speed and efficiency of satellite–terrestrial data analysis.

2.3. Framework of the SAGIN model

SAGIN offers a broader perspective by integrating space, air, and ground layers. Its architecture consists of three layers based on the spatial positions of the components, as shown in Fig. 3.

(1) **Space layer.** This layer includes LEO, medium-Earth-orbit (MEO), and geostationary-Earth-orbit (GEO) satellites, among others. LEO satellites, which are positioned closer to Earth, provide low-latency communication services and are ideal for real-time applications. MEO satellites, which are situated at medium altitudes, are commonly used for navigation systems such as the global positioning system, offering a balance between coverage and latency. GEO satellites, which are positioned at high altitudes, remain fixed relative to the surface of the Earth and deliver continuous communication services over large areas. These satellites establish links using free space optics, ensuring efficient data transmission and information exchange across the network. Together, they form a comprehensive system supporting global communication, navigation, and observation services.

(2) **Air layer.** This layer, which includes UAVs, high-altitude balloons, and high-altitude platforms, primarily handles data relay and transmission. It supports a range of applications such as collecting environmental data and transmitting high-resolution imagery. These platforms are essential for tasks such as disaster response, atmospheric data collection, and geospatial mapping.

(3) **Ground layer.** This layer includes diverse networks such as ground infrastructure networks, vehicular *ad-hoc* networks, and industrial Internet of Things (IoT) networks. This key layer lever-

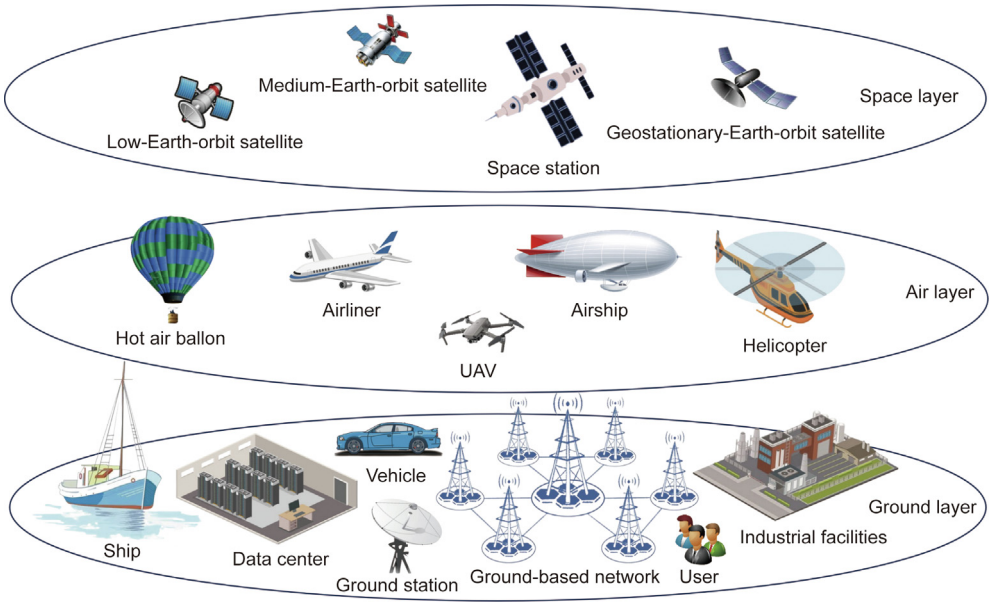


Fig. 3. The three-tier architecture of SAGIN.

ages mature technologies and stable links to ensure consistent connectivity and information exchange. Its primary functions include data collection, processing, and user services, supporting applications such as autonomous vehicles and industrial automation. Furthermore, this layer is extensively used in 5G networks, excelling in high-speed data transmission, low-latency communication, and large-scale device connectivity.

3. Problem statement

This section outlines the challenges associated with satellite–terrestrial integration. The discussion includes an exploration and categorization of common issues in both terrestrial and satellite networks. In addition, the main challenges are summarized and categorized to provide an overview of satellite–terrestrial integration issues.

3.1. Problem formulation in terrestrial networks

The STIN requires researchers to address the inherent issues of terrestrial networks as well as the new challenges that arise. Terrestrial networks present several challenges, such as end-to-end security and energy consumption. Table 2 [23–36] summarizes relevant studies in this field, along with their objectives and limitations.

3.1.1. Fundamental performance of terrestrial networks

The fundamental performance of terrestrial networks directly impacts the quality of experience (QoE). Specifically, this performance encompasses various aspects such as efficiency, latency, and security. Although these individual elements offer different perspectives, the goal is to provide users with essential communication services.

(1) **Spectrum management.** Efficient spectrum management and allocation have received widespread attention as a means of optimizing performance in modern wireless networks. Patil et al. [23] investigated intelligent spectrum sharing using AI and machine learning techniques. Their research aimed to tackle the challenges of spectrum allocation and enhance system performance through efficient management. Similarly, Ullah et al. [24] explored coded massive multiple-input multiple-output with orthogonal time frequency and space modulation to increase spectral efficiency. However, significant challenges in spectrum management and allocation persist, highlighting the need for further advancements in this area.

(2) **Throughput and latency.** To achieve efficient wireless communication, it is essential to increase throughput and reduce latency. Effective management of these factors is critical for modern applications, such as real-time video streaming and online gaming. The coexistence problem between enhanced mobile broadband (eMBB) and ultra-reliable low-latency communication (URLLC) has been investigated [25], with a focus on resource scheduling. The researchers outlined five key strategies: multiplexing, machine learning, network slicing, cloud radio access network architecture, and quality of service (QoS) metrics. Each strategy was assessed based on its effectiveness in balancing eMBB throughput with the latency and reliability requirements of URLLC. Mourtzis et al. [26] explored how 5G and the tactile Internet can address latency issues and enhance smart manufacturing. Current challenges include effectively managing the coexistence of eMBB and URLLC by optimizing throughput while satisfying stringent latency and reliability requirements.

(3) **Security.** Security presents significant challenges regarding safeguarding sensitive information and ensuring secure communication. Addressing these challenges requires robust measures to protect data integrity and maintain user confidentiality. Chawla and Mehra [27] analyzed security challenges in 5G-enabled IoT, focusing on open web security and mutual authentication. Their study highlighted limitations in classical cryptographic methods, emphasizing the need for quantum-resistant solutions. Similarly, Ramezanpour and Jagannath [28] pointed out the critical need for zero-trust principles in 5G/6G networks. They focused on challenges related to untrusted network components and proposed an intelligent zero-trust architecture. Their work emphasizes the need for machine-learning-based components to increase security in dynamic and untrusted network environments. Furthermore, Afaq et al. [29] identified significant security issues in 5G networks, such as identity tampering and supply-chain poisoning, emphasizing

Table 2
Problem formulation for terrestrial networks.

Focus	Type	Ref.	Objectives	Methods	Restrictions
Fundamental performance	Spectrum management	[23]	Efficient spectrum allocation	Spectrum utilization scheme introducing AI and machine learning	Spectrum management
		[24]	Optimize spectral efficiency and reduce channel estimation errors	Orthogonal time–frequency space-modulation method	Bit error rate
	Throughput and latency	[25]	Increase enhanced mobile broadband throughput	Mechanisms for enhanced mobile broadband and ultra-reliable low-latency communication coexistence	Stringent latency and reliability requirements for ultra-reliable low-latency communication services
		[26]	Improve backbone reliability and reduce latency	Introducing the tactile Internet	Communication delay
	Security	[27]	End-to-end security	Quantum-based solutions	Open web security + mutual authentication
		[28]	Provide information security in an untrusted network	Dynamic trust algorithm based on AI and the zero-trust principle	Connecting more devices increases the attack surface
		[29]	Improve dynamics of security mechanisms	Automation architecture based on machine learning	Security challenges + privacy issues
	Automation and intelligence	[30]	Orchestration opimization	Distributed automation solutions driven by AI	Delay + reliability + availability
		[31]	Robust interference management and resource scheduling	Cooperative resource allocation with spectrum sharing	Communication resources limited
	Caching and content distribution	[32]	Maintain quality of service and quality of experience for users and service providers	Device-to-device sharing based on cached most popular content	Access delay
		[33]	Enable efficient device-to-device communication	Distributed access control mechanism based on blockchain	Single point of failure + high computing overhead
Future development direction	Energy efficiency and sustainability	[34]	Promote sustainability and reduce operational costs	Use of renewable energy and management technologies	System energy consumption + CO_2 emissions
		[35]	Minimize energy consumption	Introducing cognition-based green communication technologies	System energy consumption + resource scarcity
	Network virtualization and flexibility	[36]	Fine-grained support for a range of heterogeneous 5G use cases	Decouple network functions for software-based virtualization	Flexibility + reconfigurable + programmable

the need for machine learning solutions to address these vulnerabilities.

(4) **Automation and intelligence.** To address performance challenges in terrestrial networks, such as latency, reliability, and availability, considerable attention has focused on automation and intelligence. Velasquez et al. [30] described several challenges in orchestrating 5G/6G networks, with a particular focus on strict latency, reliability, and availability requirements. They proposed an AI-driven orchestration solution, aiming to manage the complexities and ensure efficient service deployment. Guo et al. [31] proposed a cooperative communication framework based on relay selection and power-allocation techniques. Their approach improved interference management and resource scheduling by optimizing relay deployment.

(5) **Caching and content distribution.** The caching and content distribution issue concerns the efficient management of popular content, adding strain to 5G and IoT networks. Prerna et al. [32] discussed the inefficiencies in content caching and distribution in 5G networks, focusing on the strain placed on the network infrastructure. Furthermore, Mistry et al. [33] conducted a review of blockchain applications in 5G-enabled IoT for industrial automation. They identified issues related to caching and content distribution in terrestrial networks, including challenges with centralized access control and the need for decentralized solutions. Their study provides a comparative analysis of existing methods, focusing on their effectiveness in addressing these challenges and improving security and data management.

Fundamental performance issues in terrestrial networks, such as inefficient spectrum management, can significantly reduce service quality and cause resource congestion. These issues lead to poor throughput and high latency, resulting in slow connections and frequent interruptions. Furthermore, security and privacy shortcomings expose users to data breaches, compounding the negative impact on service quality. Additionally, insufficient automation hampers network adaptability, while poor caching increases latency and delays data access.

3.1.2. Future development direction of terrestrial networks

To meet the evolving demands of wireless networks, several research domains of future developments are emerging, focusing on energy sustainability and network virtualization. The aim of energy sustainability is to optimize energy consumption and reduce environmental impact, while network virtualization enhances flexibility and resource management. However, implementation challenges persist and hinder the overall effectiveness and performance. Addressing these challenges is crucial in ensuring that these areas can effectively contribute to the advancement of wireless networks.

- **Energy sustainability.** Energy sustainability is a critical concern in the advancement of communication networks, particularly as mobile data demand rises. Several studies have examined the limitations of existing systems and the challenges of integrating renewable energy. Although efforts to reduce energy consumption and carbon emissions are underway, many studies prioritize short-term solutions and overlook long-term sustainability challenges. Israr et al. [34] emphasized that terrestrial networks present significant challenges in managing increased energy consumption, operational expenses, and carbon emissions. While renewable energy offers a promising solution, insufficient implementation continues to hinder energy sustainability in terrestrial networks. Srivastava et al. [35] suggested that the increasing mobile data demand is leading to more base stations, increasing energy usage and carbon emissions. This trend underscores the critical necessity for more sustainable and energy-efficient approaches in the development of future networks.

- **Network virtualization.** As the shift toward next-generation mobile networks advances, network virtualization is becoming increasingly vital. Despite its importance, current terrestrial networks present critical challenges in implementing this technology. For instance, Condoluci and Mahmoodi [36] highlighted the shortcomings of existing systems in adopting network virtualization. They proposed a flexible architecture to improve resource allocation and adaptability in traditional static systems. Although software-based virtualization offer significant benefits, many terrestrial networks struggle with their effective implementation. These challenges lead to suboptimal performance and limit overall adaptability.

Effective energy sustainability and network virtualization are essential for enhancing network performance beyond basic functionalities. Energy sustainability focuses on optimizing energy use and minimizing environmental impact. However, inadequate management and low integration of renewable resources often lead to higher operational costs and environmental concerns. Network virtualization aims to improve flexibility and efficiency through virtual network resources. However, many terrestrial networks present implementation challenges, leading to inefficiencies and diminished operational effectiveness. It is crucial to address these issues in order to advance network capabilities and ensure sustainable development.

3.2. Problem formulation in satellite networks

Due to their wide coverage and high altitude, satellite networks encounter persistent challenges related to resource management and communication reliability. Table 3 [37–52] categorizes these problems and outlines their solutions.

(1) **Connectivity and stability.** Stable connectivity and network stability are crucial for satellite operations because of the dynamic nature of satellite movements. The periodic orbital motions of satellites cause intermittent visibility issues, which may disrupt network topology and affect overall stability. These topology changes also require real-time resource allocation and interlayer coordination to maintain reliability in STINs. Several works have explored possible solutions to address these problems. For instance, Shayea et al. [37] proposed a mechanism to enhance connectivity and stability in LEO satellite systems. They identified limitations that impact communication reliability, even with the inherent advantages of LEO satellites. Future research directions are outlined in the paper to address these issues. Li et al. [38] proposed a mechanism to improve connectivity and stability in networks of LEO, MEO, and GEO satellites. They noted that periodic orbital motions lead to intermittent node visibility, which makes maintaining a reliable network topology more complicated and affects stability in both the spatial and temporal dimensions. To summarize, connectivity and stability are essential in satellite networks due to the challenges posed by the dynamic nature of satellite movements. Periodic orbital motions create intermittent satellite visibility, leading to frequent changes in network topologies. These changes can disrupt communication and degrade overall network performance. Solutions that increase both connectivity and stability are crucial for achieving robust performance in a triple-layer satellite network.

(2) **Security and privacy.** As satellite networks evolve, security and privacy issues are becoming increasingly critical because of the expanded attack surface and the complexity of protecting real-time data. Traditional security measures often struggle to address the characteristics of satellite communications, such as open wireless channels and dynamic network topology. The integration of satellite and terrestrial networks in STINs complicates security management, such as encryption and access control. Additionally, the constantly changing topology and real-time coordination between network layers introduce more vulnerabilities and expand the attack surface. With the development of 6G, quantum computing, and AI, new challenges are emerging in ensuring data integrity and preventing unauthorized access. These challenges include protecting sensitive data, ensuring secure communication, and adapting to changing network environments. Furthermore, the growing number of connected devices in IoT and satellite networks complicates the management of secure communication. Several studies have pointed out these challenges in satellite and IoT security. Tedeschi et al. [39] reviewed link-layer security threats and solutions, focusing on physical-layer security and cryptographic techniques. Their paper highlights key issues such as anti-jamming and anti-spoofing and outlines future research directions. Zhang et al. [40] explored security and privacy challenges in 6G systems, discussing the limitations of traditional cryptography against physical-layer attacks. They also explored how AI can improve physical-layer security by reviewing machine learning techniques to enhance 6G security. Furthermore, Javeed et al. [41] identified vulnerabilities in IoT security for 6G networks, such as challenges with device authentication and data encryption.

To summarize, the large number of IoT devices and the limited available resources introduce additional security risks, particularly with the need for real-time communication. The evolution of satellite networks introduces significant security and privacy challenges. Traditional measures often fall short in addressing these issues. The increasing number of IoT devices complicates secure communication and maintaining network stability.

(3) **Resource management and efficiency.** Efficient resource management is crucial for optimizing computing, communication, and other services in large satellite constellations. The growing complexity of satellite networks necessitates effective allocation of resources. In STINs, resource allocation becomes even more complex because of the interdependence between the satellite and terrestrial layers, which requires real-time coordination to ensure efficient system operation. These challenges arise from differences in latency, data rates, and resource demands between the two layers, and require adaptive strategies for optimal performance. Wang et al. [42] proposed an algorithm to address the satellite range scheduling problem, optimizing resource allocation in satellite operations. They highlighted the importance of efficient resource management to handle the growing number of satellites in orbit. Lin et al. [43] identified challenges in efficient transmission resource management due to high imaging data volumes and the dynamic nature of satellite networks. Wang et al. [44] discussed challenges in managing and optimizing satellite range scheduling. With more satellites and diverse needs, balancing task profits, antenna resources, and timely completion becomes more complex. To summarize, resource management is becoming increasingly complex due to the increasing number of satellites and the dynamic nature of network traffic. The growing demand for seamless global coverage and uninterrupted services further stresses the available satellite resources. Efficient resource allocation is crucial for optimizing operations, particularly with high data volumes and diverse scheduling needs. By improving task completion, balancing antenna resources, and ensuring timely operations, effective management plays a vital role in overall network performance.

(4) **Scheduling and task management.** Effective scheduling and task management are crucial for optimizing performance. Similar to how participatory sensing allows vehicles to select routes based on their needs, satellites need efficient methods for task allocation. Managing satellite observation schedules requires address-

Table 3
Problem formulation for satellite networks.

Type	Ref.	Objectives	Methods	Restrictions
Connectivity and stability	[37]	Provide reliable communication	Explaining LEO systems and services for support	Connectivity + stability + mobility
	[38]	Ensure the topological connectivity of satellite systems	Investigating security threats and challenges	Periodic movement and intermittent visibility of satellites
Security and privacy	[39]	Enhance the security of satellite communication systems	Investigating security threats and challenges	Increased attack surface + reduced security
	[40]	Achieve smarter and more secure 6G systems	Identifying open issues to inspire further research	Various security threats
	[41]	Enhance the security and privacy of IoT systems	Introducing quantum computing and federated learning	System heterogeneity + resource constraints
Resource management and efficiency	[42]	Achieve efficient resource scheduling	A fitness-based task-resource matching algorithm	Resource scarcity
	[43]	Ensure optimal utilization of interplanetary link resources	A highly effective sequential two-phased heuristic algorithm	Miniaturization + cost reduction
	[44]	Meet the diversified needs of users for satellite ranging	Differential evolution with spatial partitioning and adaptive selection	Task profit + load balancing + timeliness
Scheduling and task management	[45]	Handle real-time dynamic tasks better	An algorithm for improved task allocation and scheduling	Limited number of satellites
	[46]	Maximize the total reward in satellite distance scheduling	A deep reinforcement learning-based heuristic scheduling method	Increased number of satellites + increased customer demands
	[47]	Total profit from scheduled requests + average image quality	A knowledge-transfer-based multi-objective framework	Multi-target scenario + real-time arrival request
	[48]	Achieve efficient inter-satellite link scheduling	A data-driven parallel adaptive large neighborhood search	Number of satellites + constellation size
	[49]	Implement efficient planning for multi-satellite coordinated observation	Decomposing and allocating tasks to satellites for effective execution	Increase in on-orbit satellites
	[50]	Achieve efficient scheduling of satellites	A reinforcement-learning-based genetic algorithm	Many factors affecting electromagnetic detection
Bandwidth and transmission	[51]	Bandwidth aggregation and redundant backup	An integrated multipath network coding transmission scheme	Long delays + frequent failures + poor resource use
	[52]	Maximize utilization and ensure user QoS	Time-varying traffic demands	Time-varying traffic demands + maximizing utilization + ensuring user QoS

ing dynamic tasks and optimizing resources, with inter-satellite communication enhancing performance. In a STIN, task management must account for the interaction between satellite and terrestrial networks, with dynamic task arrivals and changing network topologies posing unique challenges.

Several studies have developed models and algorithms to address these challenges. Li et al. [45] investigated the satellite observation scheduling problem with common dynamic tasks, noting the challenges posed by uncertain task arrival times and the growing number of delayable observations. Ou et al. [46] highlighted challenges in satellite range scheduling problems as the number of satellites and client demands grows. Efficient task assignment and scheduling are key to maintaining satellite performance. Wei et al. [47] identified challenges in scheduling real-time observation requests for observation satellites, noting issues in managing dynamic task arrivals, maximizing profit, and ensuring high image quality. Liu et al. [48] investigated the challenges of inter-satellite link scheduling within large-scale low-orbit satellite constellations. The problem involves managing communication between numerous satellites to ensure efficient data transmission. The researchers identified key issues related to optimizing link schedules and minimizing transmission delays. Song et al. [49] addressed the multi-satellite joint observation planning problem, focusing on task allocation among satellites. The challenge of efficiently distributing observation tasks among multiple satellites to optimize mission performance was examined. Their study highlights the complexity of managing satellite resources and coordinating tasks to achieve effective joint observations. Song et al. [50] addressed the electromagnetic detection satellite scheduling problem, proposing the use of a mixed-integer programming model and a reinforcement-learning-based genetic algorithm. Their study emphasizes several challenges, including the need to consider detection modes, bandwidth, and other factors that impact scheduling efficiency.

To summarize, scheduling and task management are critical for optimizing performance. Significant challenges arise in managing dynamic task arrivals and coordinating inter-satellite communications. The integration of satellite and terrestrial networks in a STIN adds complexity due to differences in mobility and resource availability between layers. Real-time coordination is required to manage task allocation and synchronization, which is likely to increase the risk of delays and inefficiencies if not carefully handled. Delays or inefficiencies in scheduling and task management can adversely affect the overall effectiveness of the satellite network. Thus, it is crucial to address these issues to ensure optimal network performance and achieve mission objectives.

(5) **Bandwidth and transmission.** As satellite networks expand, effective bandwidth and transmission management become essential due to complex resource allocation. Traditional methods struggle with issues such as prolonged propagation delays and frequent connection failures, which hinder effective bandwidth management. These limitations lead to suboptimal resource utilization and negatively impact overall network performance. Furthermore, multi-beam satellites (MBSs) for 6G create challenges in bandwidth allocation and service quality, complicating resource optimization as traffic demands grow.

Several studies have highlighted the above issues, emphasizing the need for advanced strategies to increase bandwidth and transmission efficiency in satellite networks. Ouyang et al. [51] addressed the challenges in bandwidth and transmission for large-scale dynamic low-Earth satellite networks. They first identified the key issues impacting network performance, including prolonged propagation delay, frequent connection failures, and suboptimal resource utilization. Next, they analyzed how these issues affect data transmission, noting that delays slow speeds, and connection failures disrupt communication. Additionally, Ouyang et al. [51] noted that suboptimal resource utilization leads to inefficient use of available bandwidth and caching resources.

Their study identifies critical areas for improving data-transmission efficiency and reducing latency in satellite communication systems. Ma et al. [52] explored the challenges in bandwidth and transmission within future MBS networks, which are integral to SAGIN for 6G. They identified difficulties in efficiently and dynamically allocating scarce bandwidth spectrum resources while ensuring QoS for users. As an MBS network scales up, the complexity of managing time-varying traffic demands and maximizing resource utilization increases. Their study underscores the importance of addressing these issues for effective large-scale MBS communication systems.

To summarize, the efficiency of bandwidth and transmission in satellite networks significantly affects overall system performance. The integration of satellite and terrestrial networks in STINs complicates bandwidth management due to the differences in coverage and communication characteristics. Satellites provide wide coverage but have long propagation delays, while terrestrial networks deliver localized, high-speed connections. Coordinating data flow and allocating bandwidth between these layers is challenging, especially given changing traffic demands and varying link conditions in satellite networks. Data-transfer speeds and reliability can be severely impacted by prolonged propagation delays and frequent connection failures. High latency and interruptions reduce data throughput while complicating traffic management in MBS networks. Thus, it is crucial to resolve these issues effectively in order to maintain high performance and reliability in large-scale satellite networks.

3.3. Existing solutions for STIN challenges

In this subsection, we classify and summarize the key technologies applied in satellite networks to highlight their fundamental performance and future development. Technologies such as advanced modulation, error correction, and resource allocation enhance spectrum efficiency, throughput, latency, and security. Traditional methods such as spectrum utilization schemes, supported by AI and machine learning, optimize spectrum allocation. Additionally, techniques such as orthogonal time frequency and space modulation increase spectral efficiency and address bit error rate issues.

As demand grows for low-latency, high-throughput services, coexistence mechanisms are essential for managing eMBB and URLLC. Such mechanisms balance stringent latency and reliability requirements, particularly for real-time applications such as the tactile Internet. End-to-end protection is achieved through better methods, such as quantum-based solutions and dynamic trust algorithms. However, the increasing number of connected devices expands the attack surface, resulting in persistent security challenges. Automation and intelligent orchestration, powered by AI, play a key role in optimizing network reliability, availability, and resource management. Efficiency is increased while decision-making is improved in networks with many connected devices and varying data traffic. Despite delays and limited resources, AI-driven automation holds potential to enhance network reliability.

Future development requires an emphasis on sustainability, energy efficiency, and network virtualization. Renewable energy management and green communication technologies aim to cut CO_2 emissions while reducing energy consumption. Network virtualization decouples hardware from network functions, enhancing flexibility and programmability for diverse 5G use cases.

In satellite–terrestrial integration, various technologies are employed to establish a holistic system of communication, including topology maintenance, network maintenance, and orchestration transmission. Fig. 4 illustrates the relationship among these three technologies. More specifically, the time-dependent spatiotemporal models used in topology construction increase

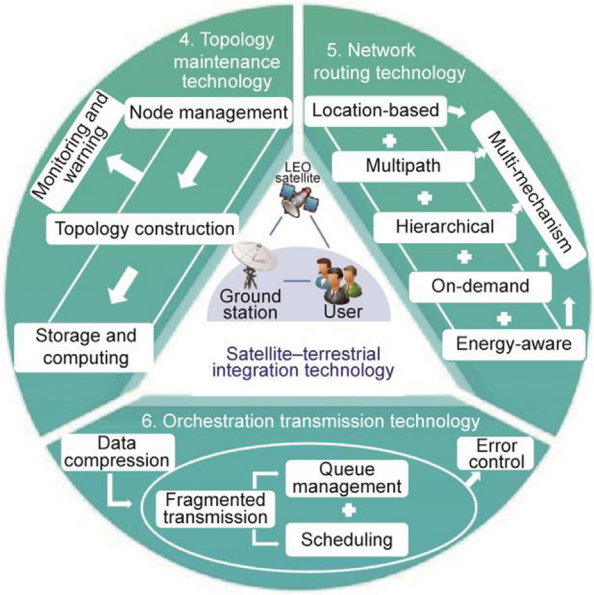


Fig. 4. Classification of key technologies in satellite–terrestrial integration.

connectivity in dynamic satellite networks, contributing to effective topology maintenance. Federated learning techniques bolster security by providing early warnings, which are crucial for ensuring network stability. Heuristic and reinforcement learning algorithms efficiently orchestrate transmissions, optimizing resource scheduling and real-time task management. The following sections explore these areas in detail, summarizing existing solutions and their limitations.

The integration of satellite and terrestrial networks allows for efficient resource sharing and load balancing, improving the utilization of available capacity, particularly during periods of high demand or congestion. When one network encounters congestion or failure, the other can seamlessly take over, ensuring continuous service and minimizing downtime. In terms of resource optimization, satellite and terrestrial networks share bandwidth dynamically, adjusting allocations based on real-time demand and network conditions. For fault recovery, the ability of the networks to complement each other enables quick rerouting of traffic through the available infrastructure, maintaining network stability. Traffic management strategies, implemented across both networks, prioritize critical data flows and optimize bandwidth usage. This collaborative approach ensures that time-sensitive applications, such as real-time communications, receive the necessary resources even during peak traffic conditions. As a result, the overall system becomes more resilient, adapting to changing conditions and maintaining optimal performance across diverse operational environments.

4. Topology maintenance technology

In this section, we summarize the research on topology management and classify these studies based on the number of layers in the satellite networks they discuss (i.e., single-layer or multi-layer satellite networks). Effective topology management in STINs requires seamless integration between the satellite and terrestrial components. However, the dynamic topology, complex network structure, and frequent interlayer collaboration—such as satellite-to-ground handovers—pose significant challenges to the development of routing efficiency and service continuity in STINs. To address these challenges, Li et al. [53] proposed a virtual overlay

network that allows for smooth interaction with the terrestrial network. This approach ensures that both network segments are coherently integrated, facilitating overall topology management. Table 4 [54–65] depicts the details of topology management, such as objectives, proposed methods, and restrictions.

4.1. Node management

Effective node management is essential in satellite networks in order to maintain the overall topology and ensure smooth network performance. Advances in node management techniques are crucial for addressing challenges such as dynamic topology changes. Researchers have developed various strategies to improve performance, stability, and efficiency in node management, ensuring optimal operation within satellite networks. These strategies often involve advanced algorithms and models tailored to specific network configurations and requirements. The key distinction is that single-layer networks focus on node placement and connectivity for topology maintenance, while multilayer networks prioritize node clustering and layer interaction optimization to improve performance.

(1) **Single-layer satellite networks.** In Ref. [54], Xu et al. proposed a topology-aware node management approach to STIN scenarios. They introduced time slot partitioning and dynamically classified satellite nodes, optimizing node roles to enhance caching efficiency and resource utilization. Vergoossen et al. [55] proposed a constellation model for trusted-node quantum key distribution (QKD) networks, with an emphasis on topology maintenance. In their work, node management was focused on using LEO QKD satellites to buffer keys with ground stations, thereby improving key distribution efficiency. Geostationary relay satellites were employed to securely transfer key combinations. Furthermore, node management was evaluated for balancing keys via inter-satellite QKD links, with a specific emphasis on maintaining the network topology under the constraints of a single-layer network.

(2) **Multilayer satellite networks.** In contrast to single-layer networks, which primarily focus on maintaining topology through methods such as node placement, multilayer networks introduce an added layer of complexity by prioritizing node clustering to enhance network performance. The challenges in multilayer networks involve optimizing the interactions between different network layers, which requires sophisticated coordination and resource allocation. Recent approaches have increasingly focused on incorporating node clustering and social attributes to further network performance. For instance, Cong et al. [56] proposed a node clustering method based on satellite interactions to improve routing performance. They leveraged the social attributes of the satellites to enhance message delivery and dynamically adjust to changing network conditions. This clustering technique considers both spatial and social dynamics, increasing the overall efficiency in multilayer network configurations.

4.2. Topology construction

Topology construction defines the overall network structure, while node management focuses on maintaining individual nodes. To enhance satellite network performance and efficiency, it is crucial to optimize topology construction. As satellite networks scale and their performance demands increase, the complexity of designing and managing network topology grows. Various strategies have been designed to improve topology construction, with a focus on efficiency, stability, and adaptability in satellite networks. Key approaches include advanced routing algorithms, dynamic link adjustment techniques, and adaptive network models. The key distinction is that single-layer networks focus on optimizing connectivity within one orbital layer, while multilayer networks enhance communication and traffic management across multiple layers to improve overall network performance and redundancy.

(1) **Single-layer satellite networks.** In single-layer networks, the focus is often on optimizing connectivity and ensuring the stability of the network within a single orbital layer. Wang et al. [57] focused on optimizing topology construction for satellite–terrestrial integration. They proposed a parallel simulation architecture with efficient topology partitioning. A resource assessment algorithm, a load-balancing-based partitioning algorithm, and a time slice algorithm were central to their approach, increasing simulation efficiency. These innovations achieved significant improvements in simulation performance. Korçak and Alagöz [58] investigated topology construction for Earth-fixed LEO satellite systems and proposed a multi-state virtual network (MSVN) architecture to address deficiencies in the virtual node approach. A new mathematical model for the MSVN was introduced, along with efficient handover mechanisms. Their results demonstrated

Table 4
Summary of topology maintenance technology.

Type	Multilayer	Ref.	Objectives	Methods	Restrictions
Node management	No	[54]	Enhance caching effectiveness	A hybrid caching strategy based on node classification	High-speed satellite movement + dynamic topology
	No	[55]	Secure communication with low latency	Constructing a key buffer using trusted satellite nodes	Network model coverage area
	Yes	[56]	High delivery rate + low latency + low network overhead	A routing algorithm based on node clustering	A large number of satellite nodes
Topology construction	No	[57]	Improve simulation efficiency of satellite network	Parallelism through network topology partitioning	Computer hardware capabilities + software performance
	No	[58]	Enhance satellite handover efficiency	Polymorphic virtual network architecture based on topology	High mobility of network nodes
	Yes	[59]	Maximize data transmission from LEO satellites	A partitioning method based on interlayer relationships	Short satellite–terrestrial communication time
Storage and computing	No	[60]	Provide computing services in remote areas	Deploying mobile edge computing servers	Limited battery capacity of ground terminals
	No	[61]	Enhance sensing + communication + computing	Processing sensor results at the edge or cloud via satellite	Balancing between resources and performance
	Yes	[62]	Higher task-completion rate and reward rate	A novel edge computing framework	Extensive solution space
Monitoring and warning	No	[63]	Improve the robustness of satellite networks	Local-state routing algorithm	Constellation size
	No	[64]	Traffic engineering + fast rerouting	Effectively recoverable segment routing	Rapid satellite movement
	Yes	[65]	Enhance network topology robustness	Various optimization algorithms	Scalability + resource constraints + security issues

that MSVN-based systems offer significant benefits over traditional virtual node-based systems.

(2) **Multilayer satellite networks.** In contrast to single-layer networks, multilayer networks incorporate satellites from multiple layers to enhance communication, traffic management, and redundancy. These additional layers introduce new challenges in coordinating the various orbits and ensuring effective data flow across layers. Compared with single-layer networks, multilayer networks use multiple orbits for better communication and traffic management. Chen et al. [59] focused on topology construction in satellite–terrestrial integration, introducing a three-layer network architecture combining LEO–Earth station links with GEO relay satellites to increase data-transmission efficiency. Their approach included a non-uniform time slot division method to handle dynamic topology changes. They also used a Stackelberg game model to optimize data-transmission strategies. This method significantly improves data volume and transmission efficiency compared with existing techniques.

4.3. Storage and computing

Resources play a vital role in SAGIN as they are essential for managing large volumes of data, such as computing, storage, and communication resources. Resource allocation is considered to increase the efficiency of data processing and save energy costs. In single-layer networks, resource allocation optimizes components within a single orbital layer, whereas multilayer networks involve coordination across multiple layers to optimize resources across different segments. The details of resource allocation are as follows:

(1) **Single-layer satellite networks.** In single-layer networks, resource allocation primarily focuses on optimizing individual network components within a single orbital layer to increase efficiency. Li et al. [60] studied energy minimization in LEO satellite edge computing networks, emphasizing offloading computing tasks and optimizing resources for battery-constrained ground terminals. To improve storage and computing efficiency, they proposed a multi-agent deep reinforcement learning algorithm that optimizes the transmit power, central processing unit frequency, data distribution, offloading decisions, and bandwidth management in a decentralized way. This algorithm outperforms random allocation, proximal policy optimization, and deep deterministic policy gradient in energy consumption. Additionally, Zhu et al. [61] optimized storage and computing resources in a STIN to enhance task-offloading efficiency. Tasks are distributed among local devices, edge nodes, and the cloud by combining subframe allocation with task partitioning to improve resource utilization.

(2) **Multilayer satellite networks.** Resource allocation becomes more complex in multilayer networks because of the involvement of multiple orbital layers, so coordination between layers is required to optimize resources across different segments. He et al. [62] proposed an edge computing framework to optimize storage and computing for agile satellite systems. They first filtered tasks at a central node and subsequently scheduled them at edge nodes using a constructive heuristic algorithm based on residual task density. The proposed method exhibited higher task completion and reward rates than other advanced techniques, while maintaining acceptable computing times.

4.4. Monitoring and warning

Monitoring and warning systems are essential for maintaining the stability and resilience of satellite network topologies. These systems detect potential failures, security threats, or resource overloads early, allowing timely interventions to prevent network degradation. In satellite networks, monitoring mechanisms track the status of links and the integrity of data transmission, ensuring service continuity. Real-time responses to these monitored issues are critical for maintaining stable topologies and quickly addressing problems such as targeted attacks or system failures. In single-layer networks, monitoring and warning systems primarily focus on the resilience of individual satellites and links, whereas multilayer networks require coordinated monitoring across multiple orbital layers to address the added complexity of interlayer interactions.

(1) **Single-layer satellite networks.** In single-layer networks, monitoring and warning systems primarily focus on the resilience of individual satellites and links within a single orbital layer. These systems aim to detect potential disruptions or failures in the network and react quickly to minimize impact. Xu et al. [63] examined the resilience of satellite constellation networks against attacks and identified two types of robustness: One evaluates the impacts on satellites and links, while the other addresses resource allocation considerations. Various attack strategies were tested, revealing that selective attacks were the most harmful. Larger constellations were generally shown to be more resilient but collapsed more quickly under a targeted attack. These results highlight the critical need for effective monitoring and warning mechanisms to strengthen network resilience. Zhang et al. [64] addressed monitoring and warning in LEO constellations using segment routing. Their delay-bounded traffic-splitting algorithm optimizes load balancing and improves failure recovery using a loop-free alternate.

(2) **Multilayer satellite networks.** In multilayer networks, the complexity of monitoring and warning systems increases due to the involvement of multiple orbital layers. Thus, such networks require a coordinated approach across layers, involving both inter-layer communication and integrated monitoring mechanisms to address the compounded challenges. Changazi et al. [65] reviewed optimization techniques for enhancing topology robustness in IoT systems. They addressed the complexities of interconnected devices, discussing algorithms such as particle swarm optimization and genetic algorithms. The scholars identified key challenges, including scalability and security, while pointing out the need for standardization. They proposed ways to improve monitoring and warning systems in smart cities and healthcare, emphasizing resilient network topologies.

5. Network routing technology

In this section, we summarize recent research on network routing technology, classifying the literature based on its focus on multilayer satellite networks. The development of STINs emphasizes the seamless interaction between the satellite and terrestrial layers, in which data upload and download between satellites and user equipment (UE) play a crucial role [66]. The coordination and optimal utilization of satellite constellations and terrestrial networks can effectively ensure routing and task delivery, adapting to dynamic mobility and intermittent connections [67]. In particular, efficient cooperation between layers is essential to overcome the complexity challenges posed by the incorporation of additional ground stations and the various limitations associated with meeting satellite service quality requirements [68]. Table 5 [69–85] provides an overview of various routing technologies, highlighting their objectives, methods, and restrictions. The following subsections present specific categories of routing techniques.

5.1. Location-based routing

In the field of location-based technologies, the focus of routing technique research differs between single-layer and multilayer satellite networks: Studies on single-layer networks prioritize optimizing adaptability and minimizing transportation costs within a single orbital layer, whereas research on multilayer networks

Table 5
Summary of network routing technology.

Type	Multilayer	Ref.	Objectives	Methods	Restrictions
Location-based	No	[69]	Improve efficiency + reduce operational costs	Adjusting satellites based on customer location and demand	Expanded distribution range + urban development
	No	[70]	Minimize total costs	A heuristic method that considers satellite storage locations	Multiple trips + multiple visits + synchronized loading
	No	[71]	Reduce overall routing costs	A study on the impact of instance parameters on global costs	Parameters including satellite positions
	Yes	[72]	Reduce management overhead and latency	A solution for location management technology	Dual mobility of satellites and users
Multipath	No	[73]	Improve transmission efficiency	A multipath routing strategy based on software-defined networking	Network congestion + transmission delay
	No	[74]	Improve load balancing + enhance QoS	A multipath routing algorithm	Network scale expansion
	No	[75]	Reduce energy consumption	A computational dependency multipath routing paradigm	Latency
	Yes	[76]	Reduce the number of interlayer link switches	A multipath routing algorithm based on ant colony optimization	QoS requirements
Hierarchical	No	[77]	Promote the forwarding efficiency of constellations	An information-centric routing scheme	Satellite mobility
	No	[78]	Reduce packet loss + lower latency	A load-aware hierarchical routing protocol	Topology dynamics + continuous traffic variations
	Yes	[79]	Meet diverse QoS requirements	A local information-based multi-objective reinforcement learning strategy	Ultra-dense + high-dynamic + large-scale
On-demand	No	[80]	Provide high-performance bandwidth services	A new paradigm called “demand islands”	Transient high-rate changes in network topology
	No	[81]	Help data routing from the source to the destination	A location-assisted on-demand routing protocol	Variations in link costs
	Yes	[82]	Reduce convergence time and end-to-end latency	An on-demand routing-based protocol	Search area + computational cost
Energy-aware	No	[83]	Balance energy usage	An energy-aware routing algorithm	Service delay requirements
	No	[84]	Avoid excessive discharge of satellite batteries	Adding energy routing planning and energy routing methods	Battery charge and discharge amount
	Yes	[85]	Maximize network energy efficiency and user acceptance	An efficient heuristic method	Complex traffic load + QoS requirements

emphasizes efficient satellite handovers and minimizing signaling overhead across multiple orbital layers. These differences are detailed in the following sections.

(1) **Single-layer satellite networks.** Tian and Hu [69] introduced a two-echelon location routing problem with recommended satellites. They developed a robust branch-and-price algorithm to improve network adaptability in city logistics. An [70] proposed a mixed-integer linear programming model for the closed-open truck and loader routing problem in biomass transportation. A constructive heuristics-based method was also developed to minimize transportation costs from satellite storage locations to a bioenergy plant. Crainic et al. [71] proposed and validated a two-echelon vehicle routing problem framework, which integrates location-based analysis for satellite placement. Satellite location rules and customer distribution are employed in the framework to optimize routing and minimize global distribution costs.

(2) **Multilayer satellite networks.** In multilayer networks, recent advancements have led to improved methods for managing location and communication across various satellite layers. The routing focus has now shifted to addressing the complexities of multiple orbital layers, which requires more sophisticated techniques for efficient satellite handovers and reduced signaling overhead. Du et al. [72] studied a dual-location area-based distributed location management method for hybrid LEO/MEO mega-satellite networks. A location-based approach was proposed to optimize satellite handovers and reduce signaling overhead, thus improving network efficiency. This method addresses the challenges of coordinating handovers across multiple satellite layers to improve communication efficiency.

5.2. Multipath routing

Multipath routing increases data transmission efficiency and reliability. It is especially important in satellite networks presenting challenges such as dynamic topologies and varying communication requirements. By utilizing multiple paths, networks can improve load balancing and ensure QoS, which makes them more resilient to failures. In single-layer networks, the focus lies on optimizing resource usage and minimizing delays; in multilayer networks, the emphasis shifts to managing the complexities of interlayer communication and routing optimization.

(1) **Single-layer satellite networks.** Qi et al. [73] proposed a dynamic multipath routing strategy using software-defined networking (SDN) for satellite networks. The strategy considers delay, bandwidth, and node load, enabling real-time route adjustments, increasing efficiency, and balancing bandwidth consumption. Based on advancements in LEO satellite networks, Liu et al. [74] introduced a multipath routing algorithm that optimizes QoS and load balancing. It utilizes an improved clustering structure to efficiently distribute traffic and minimize delays. Wang et al. [75] introduced a computing-dependent multipath routing (CDMR) paradigm for STINs. This method optimizes energy consumption by mapping tasks to multipath routes while ensuring the processing of latency constraints. The challenges in single-layer networks are primarily related to optimizing resource usage and minimizing delays within a more constrained topology.

(2) **Multilayer satellite networks.** Yang and Yao [76] proposed a multipath routing algorithm using ant colony optimization for software-defined satellite networks (SDSNs). They optimized the interlayer link handoff algorithm to reduce handoffs. By leveraging global topology information, their approach calculates multiple distinct paths, enhancing data-transmission choices and improving network security. The complexity in multilayer networks arises from managing the interactions between different satellite layers and optimizing routing across these layers.

5.3. Hierarchical routing

Hierarchical routing in STINs leverages a two-layer architecture, with satellites and the ground network being organized into

separate layers. In a STIN, routing is optimized within each layer: Satellites handle inter-satellite and satellite-to-ground communication, while the ground network manages user connections. In single-layer networks, the focus is on optimizing routing within a relatively homogeneous environment; in multilayer networks, the emphasis shifts to addressing interlayer communication complexities and diverse service requirements.

(1) **Single-layer satellite networks.** Yan et al. [77] proposed a logic-path-identified hierarchical routing method for large-scale LEO satellite networks. This approach partitions the constellation into groups and designs a path identifier for efficient routing, leading to significant improvements in packet delivery ratios. Similarly, a load-aware hierarchical information-centric routing protocol was proposed [78] to enhance routing in large-scale LEO satellite networks. The protocol effectively addresses topology dynamics and traffic variations. In single-layer networks, the focus is on optimizing resource allocation and ensuring efficiency despite the limited variety of available communication paths.

(2) **Multilayer satellite networks.** The challenges here stem from the complex interactions between different satellite layers, which make it difficult to ensure seamless communication across diverse altitudes and communication capabilities. A hierarchical routing strategy was proposed [79] for ultra-dense free-space optical LEO satellite networks. It incorporates a dual-layer network architecture, combining MEO and LEO satellites. Regional network divisions are employed to increase routing efficiency and reduce complexity. In addition, a multi-objective reinforcement-learning-based routing strategy is introduced to address the differentiated QoS requirements for various terrestrial applications. In multilayer networks, the focus shifts to managing interlayer communication, optimizing routing across heterogeneous satellites, and meeting the diverse QoS needs of multiple services.

5.4. On-demand routing

In this subsection, we explore on-demand routing strategies in satellite networks, emphasizing the adaptability of these strategies in response to changing topologies. These strategies are crucial for enhancing data transmission due to the diverse and dynamic nature of satellite configurations. On-demand routing optimizes the adaptation of data-transmission paths to dynamic environments with frequent satellite movements and variable connectivity. Recent research indicates that on-demand methods can lower latency and boost throughput by adapting routes to network conditions. In single-layer networks, the focus is on efficient routing within a relatively simple and homogeneous environment; in multilayer networks, the emphasis lies on managing interlayer communication and adapting to the heterogeneity of satellite layers.

(1) **Single-layer satellite networks.** In single-layer satellite networks, the routing strategies focus on adjusting to dynamic satellite topologies within a relatively homogeneous environment. These networks primarily deal with challenges such as handling frequent satellite movements and ensuring efficient routing within a relatively simple single-satellite layer topology. Markovitz and Segal [80] introduced the demand island routing method for LEO satellite networks. In this method, the service area is divided into various geographical regions, each with a gateway at its center. Satellites cover all regions to control the computing cost (i.e., $O(1)$) for routing. This method supports real-time multipath routing while managing bandwidth, latency, and satellite handovers efficiently. Papapetrou et al. [81] proposed a location-assisted on-demand routing protocol tailored for LEO satellite networks. Leveraging on-demand routing, which has not been previously explored in satellite networks, the protocol aims to optimize data transmission. The protocol was evaluated in terms of various link-cost metrics and exhibited superior performance over centralized approaches in simulations. In single-layer networks, the focus is on optimizing routing efficiency while managing the relatively limited complexity and path options.

(2) **Multilayer satellite networks.** In these networks, the challenges include managing interlayer routing, optimizing the data flow between satellites in different orbits, and accounting for the varying communication capabilities of each layer. Ji et al. [82] proposed an A-Star algorithm-based on-demand routing (ASOR) protocol for hierarchical LEO and MEO satellite networks. The protocol employs an improved A-Star algorithm to reduce the search area and computational costs. By calculating the optimal path on-demand, ASOR ensures optimal routes without considering broken nodes. The simulation results showed that ASOR outperformed the shortest global route protocol in terms of convergence time and end-to-end delay. In multilayer networks, the focus is on optimizing inter-satellite communication across heterogeneous layers and ensuring that the routing is adaptable to both topological changes and varying inter-satellite link qualities.

5.5. Energy-aware routing

Here, we summarize relevant works on energy-aware routing strategies in satellite networks. LEO satellites present the challenges of limited battery capacity and uneven traffic distribution. To address these issues, the goals are to extend the lifetime of satellites, reduce energy use, and ensure network performance. Energy-aware routing methods optimize power consumption while maintaining service quality by considering solar panel output, Earth's obstruction, and traffic demands. Recent studies have targeted energy-efficient resource allocation in multilayer satellite networks to optimize traffic and energy use. In single-layer networks, the focus is on reducing energy consumption and balancing traffic within a uniform satellite layer; in multilayer networks, the emphasis lies on coordinating energy use and optimizing traffic across heterogeneous layers.

(1) **Single-layer satellite networks.** The main challenges in these networks are managing limited energy resources and balancing traffic distribution among satellites. Yang et al. [83] examined energy-depletion challenges in LEO satellite networks caused by limited battery capacity and uneven traffic distribution. An available energy routing algorithm was proposed, modeling satellite energy based on solar panel output and energy consumption due to shadows cast by the Earth. The algorithm selects routing paths based on service delay requirements, aiming to reduce energy consumption while maintaining QoS. The numerical results showed a 12.53% reduction in energy consumption and a 23.72% decrease in packet loss. Macambira et al. [84] examined energy-efficient routing in LEO satellite networks in order to extend satellite lifetime. They proposed two routing methods that optimize traffic to reduce the battery depth of discharge (DOD). These methods, denoted as energy routing prUning (ERU)-DOD and energy routing penAlty (ERA)-DOD, prevent over-discharging by pruning or penalizing links when battery levels drop too low.

(2) **Multilayer satellite networks.** In these networks, the challenges include coordinating energy use across different satellite layers, optimizing traffic routing, and managing interlayer communication. Energy-efficient resource allocation is essential in multilayer satellite networks. Most of the research in this area focuses on optimizing user association, traffic routing, and x-Network function placement in order to enhance overall performance. Mesodiakaki et al. [85] addressed these challenges by formulating the problem as a mixed-integer linear program with the goal of maximizing energy efficiency and user acceptance ratios while managing complex network constraints.

6. Orchestration transmission technology

In this section, we summarize relevant research on orchestration transmission, highlighting differences in focus in research on single-layer versus multilayer satellite networks. Single-layer studies primarily optimize routing and scheduling, addressing data compression and queue management for improved efficiency. In contrast, multilayer research explores cross-layer interactions, emphasizing resource allocation and error control across diverse satellite configurations. As discussed in Ref. [86], effective resource orchestration through the cooperation of both layers is essential to guarantee service delivery, especially for ultra-remote real-time applications with long transmission distances and high delay requirements. The satellite network performs necessary on-board processing before data is downloaded to the terrestrial network for further applications, ultimately ensuring continuous service support. More specifically, due to the network heterogeneity and the mobility of internal nodes, effective resource orchestration for service delivery presents numerous challenges. Table 6 [43,87–101] summarizes the literature, providing a clearer view of potential solutions to these challenges in orchestration transmission.

6.1. Data compression

In this subsection, we explore research related to data compression in satellite networks, particularly regarding data-transmission orchestration. Effective data compression increases transmission efficiency while reducing bandwidth usage. In single-layer networks, the focus is on routing and resource use within a uniform structure, while multilayer networks require a focus on interlayer interactions and seamless data flow.

(1) **Single-layer satellite networks.** The primary challenges in single-layer networks include managing the variability of energy and buffer capacities across satellites and optimizing data transfer rates. Wang and Chin [87] proposed a mixed-integer linear program to minimize the download time for data from small satellites to ground stations. Specifically, the model tackles challenges related to varying energy and buffer constraints. It utilizes the technology of routing, link scheduling, and data aggregation rates, achieving a total download time reduction of 52%–84% through network data aggregation. Similarly, Zhang et al. [88] proposed a data acquisition and monitoring scheme for satellite networks, utilizing extended Berkeley packet filter technology for real-time data collection. The max–min fairness algorithm was employed to optimize routing, resulting in significant improvements in data-transmission efficiency. In single-layer networks, the focus is on optimizing routing and resource allocation within a uniform satellite structure.

(2) **Multilayer satellite networks.** The primary challenges in multilayer networks involve managing the interaction between various satellite layers, optimizing data-transmission paths across layers, and ensuring seamless data flow between satellites. Parkes et al. [89] introduced the Hi-SIDE project to develop satellite data chain technologies for future Earth observation and telecommunication systems. They emphasized the implementation of SpaceFibre to interconnect the data chain elements across satellite configurations, improving data-transmission capabilities.

6.2. Fragmented transmission

Fragmented transmission addresses the challenges of transmitting large data packets in satellite networks by dividing them into smaller fragments. Unlike traditional segmentation, which primar-

Table 6
Summary of orchestration transmission technology.

Type	Multilayer	Ref.	Objectives	Methods	Restrictions
Data compression	No	[87]	Minimize satellite-to-ground station download time	Introducing mixed-integer linear programming	Different satellite energy + buffering constraints
	No	[88]	Improve satellite bandwidth utilization	A novel data collection and monitoring scheme	Time-varying characteristics
	Yes	[89]	Increase efficiency and reliability of data transmission	Data link elements connect via the SpaceFibre	Demand for next-generation data-intensive missions
Fragmented transmission	No	[90]	Reduce end-to-end latency	A fragment forwarding method in named data networks	Limited maximum transmission unit of the link
	No	[91]	Improve transmission control protocol performance on GEO satellites	A segment-based transmission protocol with majority decoding	Severe damage to ground communication systems
	Yes	[92]	Improve handover success rate + reduce handover frequency	A method for batch ephemeris updates and fast handovers	Signaling overhead
Queue management	No	[93]	Increase packet transmission success	Markov chain model considering queue length	Airborne resources
	No	[94]	Reduce latency between source and destination	Comparing delays in retransmission and encoding	Packet arrival distribution
	Yes	[95]	Investigate the relationship between link inactivity and buffer status	Model the acquire–store–forward process as a queue	Intermittent connections of satellite links
	Yes	[96]	Accurately evaluate satellite network performance	Queueing analysis of SDSNs	Coarse control + long delays + low efficiency
Scheduling	No	[43]	Ensure data integrity	A new heuristic method called mission residual volume factor	Miniaturization + cost
	No	[97]	Improve satellite integration scheduling efficiency	A model considering data transmission	Limited satellite data storage capacity
	Yes	[98]	Improve work efficiency and task completion rate	A model considering data breakpoint transmission	Growing data transmission demand
Error control	No	[99]	Reduce positioning error caused by the simulator	A delay parameter-leading simulation control method	Time-varying parameters of the transmission channel
	No	[100]	Minimize wireless errors on transmission control protocol performance	A threshold-based transmission control protocol	Misinterpreted as congestion loss
	Yes	[101]	Lower the packet error rate	Error model used for simulation purposes	Interaction among coding, framing, and reconstruction

ily addresses bandwidth limitations, this approach is tailored to the unique characteristics of satellite networks, such as high latency and dynamic topology. By breaking data into smaller units, fragmented transmission reduces packet loss, increases bandwidth efficiency, and ensures reliable data delivery in complex, variable network environments. In single-layer networks, the focus is on minimizing delays and improving protocol efficiency; in multilayer networks, the focus shifts to integrating different satellite types and optimizing handovers to reduce delays.

(1) **Single-layer satellite networks.** In single-layer satellite networks, challenges include minimizing forwarding delays and addressing performance degradation in transport protocols under high-latency conditions. The focus is on improving protocol efficiency and data handling within the constraints of a single satellite type. Diao et al. [90] investigated low-delay fragmented transmission in LEO satellite networks to underpin large packet transmission. They proposed the direct forwarding and reuse of fragments method. This approach increases transmission efficiency by reducing forwarding delays and allows for the effective reuse of cached fragments. The approach contrasts with traditional transmission methods, which often suffer from retransmission delays. Wiedmeier and Tyrer [91] proposed a novel transport layer protocol for GEO satellites that supports reliable data transmission. This method utilizes multiple segment transmission with majority decoding to tackle transmission control protocol (TCP) performance degradation. These fragmented transmission methods improve reliability and reduce delays by minimizing packet loss, optimizing bandwidth, and adapting to the high latency and dynamic conditions of satellite networks.

(2) **Multilayer satellite networks.** In multilayer networks, the design considerations are focused on achieving seamless integration between different satellite types while addressing the signaling and handover complexities inherent in heterogeneous systems. Huang and Xia [92] proposed a satellite networking approach for batch ephemeris updates and fast handovers. They organized satellites into clusters with centralized controllers to reduce satellite–ground interactions and signaling overhead. This allows UE to make handover decisions using centralized ephemeris data. As a result, the UE does not need to send measurement reports or receive handover commands, which improves the success rates and minimizes interactions. This approach improves handover success rates and reduces delays compared with traditional methods that rely on real-time updates.

6.3. Queue management

Queue management involves dynamically adjusting queue lengths based on real-time traffic to minimize delays and enhance throughput. The challenges, design considerations, and technological approaches differ significantly between single-layer and multilayer satellite networks due to variations in their network structures and operational complexities. In single-layer networks, the focus is on managing congestion and minimizing delays; in multilayer networks, the focus shifts to handling interlayer interactions and optimizing queue management across diverse satellite types.

(1) **Single-layer satellite networks.** In single-layer satellite networks, the challenges include managing congestion and minimizing the delays caused by high traffic loads. Feng et al. [93] proposed a queue management framework under beam hopping by modeling user queue dynamics with Markov chains. They showed that deterministic beam hopping achieves lower user delay near saturation compared with probabilistic strategies. Barbero and Ytrehus [94] examined protocol delays in bidirectional unicast communication over satellite channels. Their findings indicate the impact of packet encoding on communication efficiency. These solutions highlight the need to optimize packet management under relatively uniform operational conditions.

(2) **Multilayer satellite networks.** In multilayer satellite networks, the challenges include managing intermittent links, coordinating interactions between satellites at different altitudes, and addressing delays caused by interlayer data forwarding. Zhu et al. [95] analyzed queue management in intermittent satellite links using a time-limited M/G/1 model. This approach overcomes traditional queuing limitations and provides insights into queue dynamics in orchestration transmission. Li et al. [96] proposed an analytical model for SDSNs to enhance queue management in orchestration transmission. They addressed inefficiencies in traditional satellite communication by incorporating OpenFlow. The scholars utilized Jackson's theorem to model interactions between the controller and forwarding nodes, as well as the store-and-forward process. These approaches underscore the focus on managing the complexities of interlayer interactions and ensuring efficient queue dynamics across diverse satellite types.

6.4. Scheduling

Effective scheduling methods ensure that satellite resources are utilized efficiently while meeting the demands for timely data transmission. These methods often involve sophisticated algorithms and models that can dynamically adapt to varying operational conditions. In single-layer networks, the focus is on optimizing resource allocation and task prioritization within a homogeneous network; in multilayer networks, the emphasis is on coordinating tasks and managing interlayer communication delays.

(1) **Single-layer satellite networks.** In single-layer satellite networks, the primary challenges include managing time-sensitive tasks and efficiently utilizing limited resources in a homogeneous environment. Lin et al. [43] proposed a scheduling approach for large-volume LEO satellite imaging data using a time-evolving graph model. The algorithm has two phases: The first optimizes the contact plan, and the second employs linear programming to create the transmission schedule. Wen et al. [97] focused on integrated scheduling for observation and transmission tasks in Earth-observation satellites. Their time-continuous model incorporates both tasks and employs a hybrid actor-critic reinforcement learning method for efficient scheduling. These methods demonstrate the focus on balancing task prioritization and resource allocation in a relatively straightforward network topology.

(2) **Multilayer satellite networks.** In multilayer satellite networks, key challenges include coordinating tasks across layers, managing interlayer communication delays, and optimizing resource allocation in a dynamic environment. Given the increasing complexity of space-based data transmission, efficient task scheduling is essential. Chen et al. [98] proposed a task scheduling method for tracking and data relay satellites. This method addresses growing demands for space-based data transmission. To increase scheduling efficiency, the scholars defined a task scheduling model and developed a two-stage method for generating a high-quality initial solution. Their method enhances the initial solution and avoids local optima, boosting task completion and resource utilization. Such solutions highlight the need to address the complexity of interlayer interactions and ensure seamless task execution in a multilayer context.

6.5. Error control

The field of error control has garnered significant attention in satellite networks to correct coding errors during data transmission. Several works have focused on coding techniques, error

detection and correction strategies, and performance optimization. In single-layer networks, error control addresses delays and performance within a homogeneous network; in multilayer networks, the focus of error control is on managing dynamic errors and interlayer interactions. A detailed description of relevant works is provided below:

(1) **Single-layer satellite networks.** In single-layer satellite networks, error control strategies primarily address the challenges arising from long propagation delays and limited bandwidth within a relatively homogeneous network. The focus is on reducing error rates while maintaining efficient resource utilization in a simpler topology. Li et al. [99] proposed a simulation control method with delay parameters in global navigation systems, in order to minimize the position gap. This improvement is crucial for ensuring reliable orchestration transmission in satellite communications. Teng and Chang [100] proposed a threshold-based TCP extension to address long round-trip delays in satellite channels. The extension prevents TCP from misinterpreting corruption losses as congestion losses. Numerous experimental results showed that the channel error was reduced thanks to the threshold-based TCP extension. These techniques emphasize addressing delay-induced issues and optimizing performance in a single-layer context.

(2) **Multilayer satellite networks.** Error control in multilayer satellite networks is more complex due to the heterogeneous nature of the network, involving interlayer data transmission and diverse operational conditions. The key challenges include managing the dynamic error patterns introduced by interlayer interactions, optical link disruptions, and varying environmental conditions. Friedrichs and Wertz [101] investigated a multilayer satellite network consisting of GEO relay satellites and LEO satellites. Broadband data from LEO satellites were transmitted via laser inter-satellite links to the GEO relay, functioning as a regenerative repeater with free space optics. To address the effects of micro vibrations on the optical link, a time-variant discrete-time model simulated error patterns. Such methods highlight the need to address the interplay between satellite layers and environmental factors in multilayer networks.

7. Application scenarios

In recent years, increased STIN applications have been seen across various domains. Here, we provide an overview of application scenarios and related research in STINs. The main scenarios are categorized into four areas: transportation, finance, industrialization, and emergency rescue.

7.1. Transportation sector

STINs provide solutions for monitoring, data offloading, resource allocation, and interconnectivity in the transportation sector, as shown in Fig. 5. In this subsection, we explore the challenges and related technologies involved in each practical application.

(1) **Monitoring management.** Remote sensing technologies play a crucial role in enhancing navigation and monitoring within transportation systems. Zang et al. [102] proposed a real-time monitoring system for dangerous goods transportation using a navigation satellite system. Their system consists of three modules for positioning, wireless communication, and information processing, which enable the real-time monitoring of vehicle locations.

(2) **Offloading management.** STINs are vital for improving caching and computation capabilities within intelligent transportation systems (ITSs). They enhance data distribution and processing by utilizing LEO satellites and high-altitude platforms. Yang et al. [103] proposed a novel approach for caching and computation offloading that leverages this integration. Their approach ensures real-time data access in remote areas, making critical information readily available for transportation applications. Specifically, they developed caching policies and a multi-agent federated learning strategy to reduce data-transmission delays.

(3) **Allocation management.** Effective resource and network allocation is crucial for optimizing the performance of ITS. Cognitive satellite–vehicular networks offer a promising framework to achieve optimized performance by addressing the complexities of resource management. Ruan et al. [104] introduced a cognitive satellite–vehicular network to enhance resource and network allocation in ITS. They explored the tradeoff between energy efficiency and spectrum efficiency and proposed a unified metric for dynamic environments. Ruan et al. [105] also explored power allocation in cognitive satellite–vehicular networks for ITS. Their energy–spectral efficiency tradeoff metric optimizes resource allocation under interference constraints.

(4) **Topology management.** The integration of terrestrial and satellite networks significantly improves communication in transportation systems, especially for IoT devices. Tello-Oquendo et al. [106] improved interconnectivity and communication in a STIN via a vertical handover framework that selects the most suitable access technology based on service requirements, enabling seamless connectivity.

7.2. Financial sector

Several studies have focused on utilizing satellite communication systems to enhance financial industry operations. Satellite networks outperform terrestrial ones in long-distance, high-capacity transmission, ensuring reliable connectivity for financial institutions. These systems facilitate crucial data transmission for timely financial decision-making and risk management. Additionally, satellite communication supports secure transactions and efficient data processing, ensuring the integrity and confidentiality of financial information.

Guskov [107] demonstrated an information satellite system that integrates geostationary and LEO satellites. The system optimizes communication channels and improves operational efficiency for financial organizations. Chen et al. [108] showed that satellite data combined with AI enhanced financial inclusivity for small farmers during agricultural disasters. Furthermore, Shaengchart and Kraiwanit [109] emphasized the importance of a satellite Internet that would expand access to financial resources and professional development opportunities in emerging economies. Satellite networks in finance enhance services and drive economic growth by connecting underserved populations.

7.3. Industrial production

Recent research has highlighted the critical link between air pollution and industrial development, emphasizing the need for effective monitoring and management. Zhang et al. [110] examined how regional air pollution impacts industrial structure upgrading. They found that a 1% increase in particulate matter lower than 2.5 μm ($PM_{2.5}$) concentration led to a 2% decrease in the height of the industrial structure, which in turn affected investment and innovation. Similarly, Fu et al. [111] used satellite observations to analyze China's industrial emissions, revealing that high NO_2 levels were mainly found in eastern regions, while higher SO_2 concentrations were located in the northern regions. Furthermore, Rojano et al. [112] assessed PM_{10} water-soluble ions in northern Colombia and found a strong link between sulfate levels and satellite data, showcasing the value of satellites for monitoring emissions.

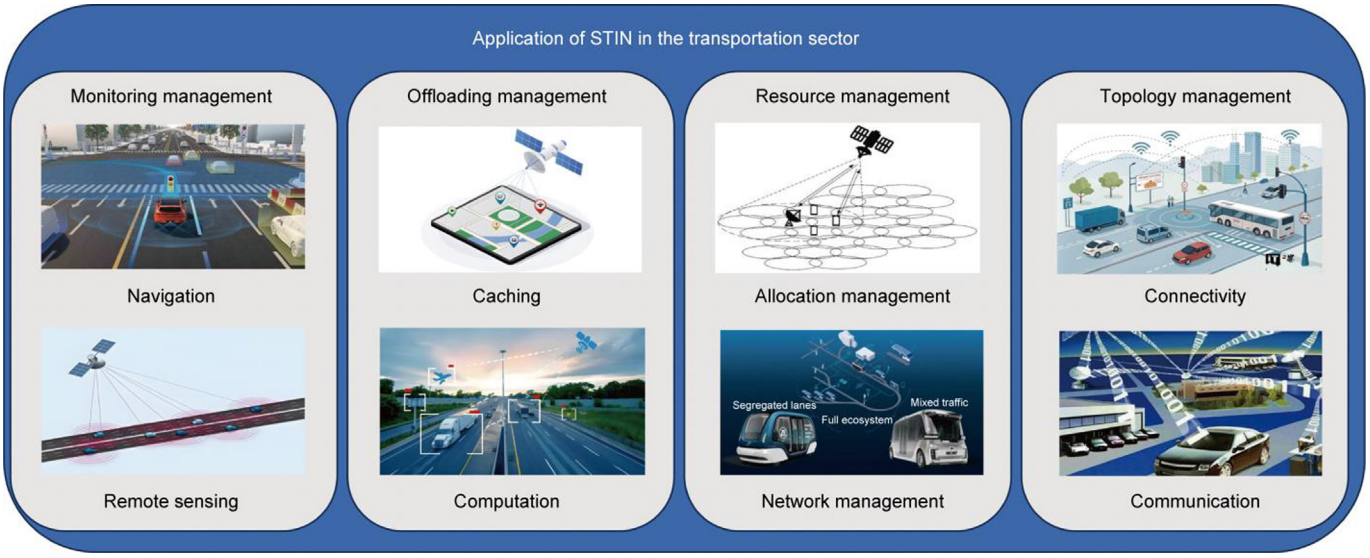


Fig. 5. Classification of STIN applications in the transportation sector.

7.4. Emergency rescue

STINs have proven to be indispensable due to their ability to overcome limitations in terrestrial communication, such as coverage gaps and signal interference. The integration of satellite communication with terrestrial networks creates robust and reliable channels for data transmission. These channels remain operational even when ground infrastructure is compromised by natural disasters such as earthquakes, floods, or typhoons.

Zheng et al. [113] demonstrated a high-throughput satellite (HTS)-based emergency rescue communication system that uses an HTS for fast communication in disaster areas, allowing quick data exchange among rescue teams and command centers to improve response efficiency. Similarly, Zheng et al. [114] highlighted the benefits of the Tiantong-1 satellite mobile communication system—China's first GEO-based mobile satellite system, which offers uninterrupted coverage, even in the most remote and disaster-impacted regions. The system offers real-time voice, data, and multimedia services, improving command during large-scale rescue operations when terrestrial networks fail. Further advancing satellite constellations in emergencies, Liu et al. [115] designed a micro-satellite constellation for all-weather remote sensing with high revisit capabilities. These advancements improve emergency response and enable real-time disaster monitoring. In addition, Liu et al. [116] reported a meteorological emergency communication system using portable satellite stations. Their system combines wireless and satellite technologies for secure communication during emergencies. These studies demonstrate how STINs enhance emergency communication systems and are thus essential for disaster relief.

8. Experimental research and datasets

In this section, we first outline the frameworks, platforms, and simulators related to STINs. Then, we discuss the relevant datasets used for simulation experiments.

8.1. Frameworks, platforms, and simulators

The framework, platform, and simulator are interconnected in a STIN, ensuring efficiency and functionality. Firstly, the framework is a high-level abstraction that organizes space–terrestrial integration activities, helping developers design applications without directly handling simulations. Secondly, the platform supports these frameworks and applications by providing essential resources and functionalities, often including simulators for modeling communication processes. The simulator executes tasks, allowing frameworks and applications to interact for simulated data or system response testing. These elements collectively create an efficient environment for the STIN, with communication networks as the foundation for their operation. We have categorized frameworks, platforms, and simulators by summarizing their common usage classifications, as shown in Table 7 [117–132].

(1) **Frameworks.** The STIN framework includes resource management, network control, data sharing, and other relevant aspects.

- DCOOL: Kim and Kwak [117] introduced DCOOL, a resource management framework designed for LEO satellite-assisted edge computing. The framework emphasizes dynamic computation offloading and resource allocation within the LEO satellite mobile edge computing architecture. By leveraging Lyapunov optimization, DCOOL aims to minimize average power consumption and propagation delay while maintaining queue stability. This approach increases workload processing efficiency and contributes to an overall power reduction in the system.
- PTMB: Li et al. [118] developed a framework based on a pre-trained Markov-based (PTMB) model, which is tailored for online satellite task scheduling in multi-task scenarios. The framework comprises two main phases: task decision-making and task allocation. PTMB employs a pre-trained Markov model and strategies for resource pre-allocation and task sequencing to improve scheduling. This framework improves task rewards and significantly decreases time consumption compared with traditional scheduling methods.
- DLSDN: Wang et al. [119] proposed a dual-layer SDN (DLSDN) framework to improve network control in LEO constellation networks. The framework integrates a scheduled update mechanism to handle predictable changes, along with a trigger-response mechanism for unexpected variations. Simulations showed that DLSDN improved network accessibility and reduced recovery times to hundreds of milliseconds.

Table 7
Frameworks, platforms, and simulators.

Type	Name	Ref.	Category	Description
Framework	DCOOL	[117]	Resource management	Dynamic computation offloading and resource allocation for LEO satellite-assisted edge computing in a ground–space integrated framework
	PTMB	[118]	Resource management	An online satellite task-scheduling framework for multi-task scenarios leveraging a pre-trained Markov-based (PTMB) model
	DLSDN	[119]	Network control	A novel dual-layer SDN (DLSDN) in-band control framework
	StarCross	[120]	Data sharing	A secure data-sharing framework for satellite IoT using redactable blockchain
Platform	CAST2000	[121]	Satellite platform	A versatile small satellite platform for high-precision control and efficient management
	DFH-3E	[122]	Satellite platform	A small GEO all-electric propulsion satellite platform
	MODIS	[123]	Satellite platform	An Earth observation platform on the Terra and Aqua satellites for multi-spectral data including a moderate-resolution imaging spectroradiometer (MODIS)
	GREAT	[124]	Software platform	The high-performance Global Navigation Satellite System Research, Application, and Teaching (GREAT) software platform
	Mini-Savi	[125]	Software platform	A simulation platform for large-scale satellite constellations, combining mininet and satellite constellation visualization
	PINOCCHIO	[126]	Software platform	The Platform Integrating Navigation and Orbital Control Capabilities Hosting Intelligent Onboard (PINOCCHIO)
	SEACOP	[127]	Software platform	The SatEllite Application Capability Open Platform (SEACOP) based on Docker and Kubernetes
Simulator	DASE	[128]	Dynamics simulator	The Dynamic Attitude Simulator Environment (DASE) for real-data-based onboard flight code simulation
	LAPAN-ITB	[129]	Dynamics simulator	A micro-satellite simulator developed by the National Institute of Aeronautics and Space together with Institut Teknologi Bandung
	GISTDA	[130]	Operations simulator	The Geo-Informatics and Space Technology Development Agency (GISTDA) simulator for validating satellite operations in development tools
	TTC-RF	[131]	Functionality simulator	The telemetry, tracking, and command–radio frequency (TTC-RF) satellite simulator emulating the radio frequency function of a transponder
	SATSIM	[132]	Functionality simulator	The MATLAB/Simulink satellite simulator for spacecraft hardware and environments

- StarCross: Du et al. [120] proposed a secure and lightweight data-sharing framework called StarCross for satellite-based IoT. The system uses a redactable blockchain to improve scalability and security. It adopts a space–ground sharing architecture, where LEO satellites act as intermediaries between different data segments. Theoretical analysis and experiments showed that StarCross significantly enhanced scalability and security compared with existing methods.

(2) **Platforms.** STIN platforms can be roughly categorized into satellite simulation platforms and application software platforms integrated with satellites.

- CAST2000: Qin et al. [121] presented the CAST2000 small satellite platform, which has been widely used in various space missions due to its high reliability and flexibility. CAST2000 supports applications such as remote sensing and communication tasks, emphasizing efficient design and resource management. Improvements in frequency design focus on addressing coupling issues to ensure better satellite response and stable on-orbit performance.
- DFH-3E: Hou et al. [122] presented the DFH-3E satellite platform, a small GEO system with all-electric propulsion. It is designed to meet medium and small communication capacity needs, supporting up to 20 transponders. The platform can be equipped with either Hall or ion electric propulsion, based on user requirements.
- MODIS: Pinna et al. [123] highlighted moderate-resolution imaging spectroradiometer (MODIS) as the most frequently utilized satellite platform for mapping and monitoring grasslands and pastures from 2000 to 2022. Despite the growing use of newer platforms such as Sentinel-2, MODIS continues to play a critical role in remote sensing due to its long-term data availability. Platforms integrated with machine learning algorithms have significantly advanced ecosystem monitoring research.
- GREAT: Li et al. [124] developed the Global Navigation Satellite System Research, Application, and Teaching (GREAT) platform for satellite geodesy and navigation. The platform focuses on space geodetic data processing, precise positioning, and orbit determination. It integrates multiple techniques for real-time geodetic parameter determination while achieving high-precision orbit and clock determination, allowing for centimeter-level positioning accuracy.
- Mini-Savi: Tang et al. [125] proposed Mini-Savi, a simulation platform for mega satellite constellations that reduces delays using inter-satellite links. The platform effectiveness was validated with the Iridium constellation and exhibited accurate satellite orbit modeling and dynamics.
- PINOCCHIO: Sabatini et al. [126] presented the Platform Integrating Navigation and Orbital Control Capabilities Hosting Intelligent Onboard (PINOCCHIO), which is characterized by its flexibility for satellite control experiments. The free-floating system simulates microgravity by reducing surface friction, enabling motion similar to orbital dynamics. Additionally, the work details the platform's navigation and communication systems, along with its guidance, navigation, and control architecture. These findings are significant for advancing free-floating platform development in both academia and industry.
- SEACOP: Wu et al. [127] introduced the SatEllite Application Capability Open Platform (SEACOP) to enhance satellite application deployment and management. The platform employs Docker and Kubernetes to provide a unified environment for satellite applications running on both ground and space nodes. Users can monitor the application status in real time through a web portal, with support for data-based, template-based, and image-based applications.

(3) **Simulators.** In STINs, the challenge of gathering extensive real-world sensing data arises from the complexity and cost asso-

ciated with large-scale deployments. Simulators use synthetic data to test the feasibility and performance of frameworks, offering insights before deployment.

- DASE: Yavuzyılmaz et al. [128] developed the Dynamic Attitude Simulator Environment (DASE) to test flight software for controlling spacecraft orientation. DASE simulates the movement and behavior of the satellite, generating realistic sensor signals to verify controllers before launch. The study presents two operational scenarios: stabilizing the satellite after launch and three-axis nadir pointing control.
- LAPAN-ITB: Triharjanto et al. [129] developed the LAPAN-ITB microsatellite simulator using MATLAB and C. The simulator supports satellite system development and operator training, reducing costs by enabling design validation without hardware. The orientation control module was tested and shown to accurately simulate satellite orientation with three orthogonal reaction wheels.
- GISTDA: Saroj et al. [130] presented the Geo-Informatics and Space Technology Development Agency (GISTDA) satellite simulator, based on SimTG, a satellite simulation tool developed by the aerospace company AIRBUS. SimTG uses a common object request broker architecture middleware for data distribution, but its complexity causes real-time performance issues. To address this, the researchers proposed using lightweight middleware, such as message-queuing telemetry transport, to reduce workloads.
- TTC-RF: Gupta et al. [131] proposed a telemetry, tracking, and command–radio frequency (TTC-RF) system simulator designed using a dedicated platform and Labview software with a software-defined radio approach. The simulator simulates all the radio frequency (RF) functionalities of a telemetry, tracking, and command (TTC) transponder, providing a configurable platform for testing and evaluation before actual satellite testing.
- SATSIM: Bodin et al. [132] introduced SATSIM, a real-time multi-satellite simulator. SATSIM validates guidance, navigation systems, and on-board software in a simulated environment. It simulates sensors, actuators, and space-craft dynamics, supporting closed-loop satellite tests.

The frameworks, platforms, and simulators used for STINs share the common goal of enhancing satellite operations. Each framework focuses on specific optimizations, such as resource management, and many have been validated through real-world testing. However, distinct differences exist within each category. For instance, each of the frameworks targets specific application domains: DCOOL emphasizes edge computing, while PTMB addresses multi-task scheduling. The platforms also support satellite operations in unique ways, categorized into simulation platforms and application software. Lastly, the simulators provide data to test these frameworks and platforms, validating their performance prior to deployment.

8.2. Datasets

Various datasets have been developed using different satellite platforms, such as the Starlink, Sentinel, and WorldView satellites. Some datasets, such as the climate-aware satellite image dataset (CASID), focus on environmental monitoring, while others concentrate on object detection and tracking. These datasets serve as essential resources for researchers to assess the effectiveness of the methods they propose, as summarized in Table 8 [133–139].

Table 8
Frameworks, platforms, and simulators.

Name	Ref.	Category	Description
WetLinks	[133]	Communication and tracking	A large and publicly available track-based Starlink measurement dataset
VehSat	[134]	Object detection	A large-scale dataset for vehicle detection in satellite imagery
SAT-MTB-SOS	[135]	Object detection	A satellite video single object segmentation benchmark dataset
iVision MRSSD	[136]	Object detection	A multi-resolution satellite based synthetic aperture radar ship detection (MRSSD) dataset
GAZADeepDav	[137]	Environmental monitoring	A high-resolution geotagged satellite image dataset for deep learning-based damage assessment via vision
S1S2-Water	[138]	Environmental monitoring	A global dataset for semantic segmentation of water bodies using Sentinel-1 and Sentinel-2 imagery
CASID	[139]	Environmental monitoring	A large-scale climate-aware satellite image dataset (CASID)

- WetLinks [133]: WetLinks is a publicly available dataset containing about 140 000 measurements from the Starlink satellite network. These measurements were collected over six months from two locations in Europe. The dataset includes key network parameters such as upload and download throughputs, round-trip time, packet loss, and traceroutes. It also incorporates data from nearby weather stations.
- VehSat [134]: This dataset can be used for vehicle detection in satellite images. It includes 4544 crops of satellite images collected from four different satellites across eight areas. The dataset contains annotations for a total of over 36 000 vehicles in various contexts and resolutions.
- SAT-MTB-SOS [135]: This dataset is a satellite video object segmentation benchmark with 113 video sequences and 13 500 frames. It covers five target categories: airplanes, trains, vehicles, ships, and buildings. The dataset includes challenges such as occlusion, motion blur, and non-rigid variations for evaluating segmentation algorithms.
- iVision MRSSD [136]: This dataset contains 11 590 synthetic aperture radar image tiles with 27 885 ship examples, each sized 512 × 512 pixels. It includes diverse scenes from six satellite sensors across the C, L, and X radar bands. The dataset is divided into training, validation, and test sets in a 70:20:10 ratio for benchmarking ship-detection algorithms.
- GAZADeepDav [137]: This dataset, which is used in conjunction with SqueezeNet to identify wartime damage, contains PlanetScope satellite images, including 7264 tiles labeled as "no damage" and 6196 tiles labeled as "damage." The accuracy of SqueezeNet has reached 98.95% and even increased to 99.10% with bidirectional long short-term memory layers.
- S1S2-Water [138]: This dataset contains 65 globally sampled Sentinel-1 and Sentinel-2 image triplets, each paired with a binary water mask. The samples are selected based on predominant landcover and the presence of water bodies. The dataset also includes metadata and a digital elevation model (DEM) from the Copernicus DEM. It is designed for training, validating, and testing convolutional neural networks for surface water body segmentation.
- CASID [139]: The CASID includes 980 satellite images, each 5000 × 5000 pixels, from 30 regions in Asia. It covers four climate zones and is designed for domain-adaptive land-cover semantic segmentation.

The datasets related to satellite technology are interconnected and collectively enhance research in various domains. Each dataset

targets specific applications, such as communication, object detection, and environmental monitoring. For instance, WetLinks provides insights into satellite communication performance, while VehSat, SAT-MTB-SOS, and iVision MRSSD focus on object detection. Environmental monitoring is covered by GAZADeepDav, S1S2-Water, and CASID, which analyze damage, water bodies, and land cover, respectively. Despite their distinct focuses, these datasets are all intended to support the common goal of advancing satellite-based research and applications.

9. Discussion and future research directions

In this section, we first discuss the impacts of STINs and relevant fields. Then, we explore development trends and potential future research directions in this domain, providing insights that highlight future development directions for STIN.

9.1. Impacts of STINs

In Sections 4–6, we categorized and summarized the three major technical approaches used to address challenges in STINs. In this subsection, we discuss the impacts of STINs on these technologies. These methods rely on STINs for communication but differ in their dependence on bandwidth, latency, and security.

(1) **Similarities.** A STIN provides essential infrastructure for data transmission between different network entities, facilitating seamless information exchange. This enables collaboration between satellites, UAVs, and ground stations, leading to efficient task execution and network operations. The shared data and information also enhance the overall system performance by enabling better cooperation and coordination across the network entities. Additionally, the performance of STINs, especially in terms of data-transmission speed and reliability, is critical to ensuring real-time responses in dynamic environments.

(2) **Differences.** The specific impacts of STINs on these methods differ based on their requirements. For topology maintenance, STINs ensure stable communication links and facilitate the dynamic adaptation of network structures, which are crucial for maintaining connectivity as network entities move. In network routing, STINs are central to transmitting large volumes of data across different layers, with low latency and high bandwidth as requirements to ensure optimal routing decisions. Finally, orchestration transmission relies on STINs to coordinate data traffic across network entities, with a focus on reliability and reducing transmission delays.

9.2. Interaction with related fields

Several areas are closely related to STINs and have contributed to STIN development in modern communication systems. In this subsection, we describe these relevant areas.

(1) **The IoT.** In a STIN, the IoT closely interacts with LEO satellites and is becoming essential for global network services in the 6G era. This integration enhances the application of IoT devices, particularly in remote areas with limited infrastructure. As the number of IoT devices increases, the need for efficient data transmission rises. STINs provide robust connectivity for IoT devices, especially in remote areas with limited infrastructure [140]. By leveraging LEO satellites, the STIN enables real-time data exchange and allows devices to adapt to changing network conditions. This collaboration improves network reliability and promotes the deployment of IoT solutions in isolated environments.

(2) **AI.** AI significantly enhances STINs by analyzing data from IoT devices and LEO satellites. AI algorithms enable real-time decision-making, optimize resource allocation, and predict network conditions, improving traffic management and communication efficiency. While challenges exist, AI integration boosts network reliability and supports the development of intelligent applications for service delivery.

(3) **Cloud and edge computing.** The integration of LEO satellites with terrestrial cloud and edge computing systems improves service delivery. Although cloud computing provides centralized processing, it may introduce latency for real-time applications. Edge computing reduces delays by bringing processing closer to the user. Combining orbital edge computing with terrestrial resources addresses connectivity challenges and enhances network performance [141].

(4) **Network security.** Cybersecurity is vital for STINs, in which data is transmitted between IoT devices and satellites. Given the increasing reliance on these networks for critical applications, robust security measures are essential to protect sensitive data and ensure uninterrupted communication. The unique architecture of a STIN, which includes multiple nodes and diverse communication channels, presents significant vulnerabilities to cyber threats. Effective cybersecurity strategies safeguard against attacks such as data breaches, unauthorized access, and denial-of-service incidents. By integrating advanced security protocols and encryption methods, STINs can maintain data integrity and user privacy, thus fostering trust and resilience in network operations.

9.3. Future research

Due to the dynamic environments in STINs, traditional communication technologies demand significant computational resources to estimate communication channels but fail to handle the frequent changes in network topology. Therefore, AI-driven solutions, such as AI-assisted strategies for RISs, federated spectrum learning, and multi-task learning, have attracted attention for improving adaptability in these dynamic environments [11–14]. Furthermore, as edge computing and AI technologies advance, conventional communication and processing methods are no longer suitable for addressing the low-latency requirements of real-time data processing in autonomous systems. Consequently, there has been increased interest in reconfigurable intelligent computational surfaces to improve safety and efficiency in edge-computing-assisted networks, especially in autonomous driving [142,143]. These developments highlight the need for further research to address the limitations of traditional methods and explore new directions, such as improving network protocol adaptability and using AI to optimize STIN performance.

10. Conclusions

In this review, we systematically examined recent research on and developments in STIN technologies in 6G. First, we introduced the background and concept of STINs, along with common technical classifications. We placed special emphasis on SAGIN, a key research model. Next, we summarized the issues mentioned in previous papers from the perspectives of both satellite and terrestrial networks. Subsequently, we provided a comprehensive overview and classification of recent STIN technologies from a unique network-layer perspective. More specifically, we categorized past research into three areas: topology maintenance, network routing, and orchestration transmission. Within each category, we performed a secondary classification based on the complexity of the architecture, distinguishing between single-layer and multilayer satellites. Additionally, we discussed mainstream application scenarios and related fields in detail. Notably,

we also summarized the relevant experimental frameworks, platforms, and simulators used in previous research.

CRediT authorship contribution statement

Nuo Chen: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Yujie Song:** Writing – review & editing, Resources, Conceptualization. **Yue Cao:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization. **Zhili Sun:** Writing – review & editing, Resources. **Bo Zhao:** Resources. **Mi Wang:** Resources. **Debiao He:** Resources. **Guojun Peng:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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